



TossMatch

Complete Project Documentation

Player application • P2P game catalog • bots • tournaments • economy • admin operations

Document version	v10.4 complete-documentation revision
Project build	TossMatch v10.4 — Balanced Coin-Use Expansion
Prepared	25 August 2026
Runtime	Self-contained local browser play-coin demonstration
Authority	Appendix V is the authoritative consolidated current specification; earlier appendices retain release history.

IMPORTANT This build is a local demonstration using virtual play coins. It is not a real-money gambling system, payment platform, secure multiplayer server, or production fairness service.

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Scope note Rules in this document describe the current HTML/JavaScript implementation, including its demo-specific behaviour. Where the UI expresses a production intention that is not fully implemented, the limitation is called out explicitly.

1. Executive summary

TossMatch v10.4 is a local browser demonstration with 33 P2P Catalog games, 20 Arcade+ modes, progression, economy, bots, Platform/Security/RG services and UX personalization. The balanced coin-use expansion adds four Catalog games, four Arcade games and four non-wager coin utilities while retaining the production deferrals from v10.3.

- All visible game modes are framed as entrant-versus-entrant. Bots may stand in for participants and can play one another without a real player.
- A waiting-room model escrows a posted stake, displays it in the queue, then auto-matches a bot where bot liquidity is enabled.
- Coin Toss and catalog matches use Web Crypto SHA-256 and retain proof material for local verification.
- Player and admin pages share the same localStorage record under the key tossmatch_v8.
- The system is intentionally local and demonstrative; it has no secure backend, account authentication, payment processing, or server-authoritative ledger.

Core product surfaces

Surface	Purpose	Primary capabilities
Player app — index.html	Player-facing experience	Play, Series, catalog games, leaderboard, player directory, lobby, shop, season, wallet, statistics, verifier
Admin panel — admin.html	Operations and economy control	Dashboard, rates, jackpot, VIP, tournaments, promotions, economy simulation, queue monitoring, bot transfers, audit and exports
Catalog reference	Design input used by the build	Defines side-, number-, pattern-, range-, multi-flip, Series, and tournament game concepts

2. Product scope and files

2.1 Deliverable files

File	Role	Dependency model
index.html	Complete player application and simulation engine	Inline CSS and JavaScript; no application framework
admin.html	Companion operations panel	Inline CSS and JavaScript; reads/writes the same localStorage state
TossMatch_Project_Documentation.docx	This project reference	Office Open XML document

2.2 Game-mode inventory

The synchronized Feature Directory contains 101 records: 90 Implemented, 3 Partial/demo-only and 8 Suggested. Player/Admin command palettes provide exact links to 53 games. New coin uses include cosmetic crafting, event tickets, non-withdrawable clan treasury contributions and private-room visual upgrades.

Area	Modes
Play	Coin Toss; public/private posting; auto-match; ×2 rematch challenge
Games — 17 catalog arenas	Over/Under, Speed Round, Tug of War, Even/Odd Sum, Closest Number, Lucky Number Battle, Sum Prediction, Higher Byte, Pattern Race, Parlay Duel, Prediction Streak, Blind Pick, Range War, Bullseye, Chain Reaction, Elimination Ladder, Mirrored Coins
Series	Bo3, Bo5, Bo7, Advantage
Tournaments	Single-flip bracket; Bo3 Series bracket

2.3 Out-of-scope for this demo

- Real-money deposits, withdrawals, custody, KYC, AML workflows, tax reporting, or payment-provider integration.
- Secure login, roles, server-side authorization, anti-cheat controls, or cryptographic keys held outside the browser.
- True networked human-versus-human matchmaking. The local demo simulates participants with bots.
- Guaranteed persistence across browsers/devices or after localStorage is cleared.
- Regulatory certification of game mathematics or RNG.

3. Architecture and state model

3.1 Runtime architecture

1. The browser loads index.html and initializes a default state if tossmatch_v8 does not exist.
2. Player actions and bot simulation mutate one in-memory state object, S.
3. The state is serialized to localStorage. At high Turbo levels, a separate four-second persistence interval limits write pressure.
4. admin.html loads the same key, merges missing defaults, renders operational views, and writes configuration changes back to localStorage.
5. The browser storage event synchronizes open player/admin tabs on the same origin.

Trust boundary The admin panel is not protected by authentication. Anyone who can open admin.html in the same browser origin can change the local demo state.

3.2 Major state domains

Domain	Representative fields	Purpose
Wallet	main, bonus, referral, rakeback, bank	Segmented player balances

Player progression	level, xp, monthWagered, streak, quests, achievements	Rewards, status, limits, and unlocks
Match queues	waiting, cups, trnys, x2room	Open escrowed entries and competitions
History	games, ledger, catalogLog, botTransfers, feed	Results, proofs, transfers, and activity
Economy	jackpot, config.house, taps, sinks	Fees, rake, sales, promotion cost, and pool accounting
Bots	99 bot profiles with balance/stats/cosmetics	Liquidity and simulated ecosystem
Configuration	rates, jackpot, VIP, rewards, features, promotions	Admin-controlled behaviour
Global statistics	heads, tails, totalGames, jackpots	RNG health and aggregate reporting

3.3 State migration behaviour

Both pages merge saved data into a current default structure. The player app adds newly seeded bots, backfills bot statistics, ensures catalog carry pools and histories can be created, and converts older bot-created Series Cups into public bot-ready Cups. Tournament format defaults to single when an older tournament lacks a format.

4. Player application and navigation

Screen	What the player can do
Play	Pick HEADS/TAILS, set stake, add taunt, post public/private bet, toggle sound/instant/auto-rebet, inspect proof, post ×2
Series	Open or join Bo3/Bo5/Bo7/Advantage Cups; view bot-ready Cups and tournaments
Games	Open one of 17 catalog game pages, choose a pick, post to its waiting room, cancel before matching, inspect result/proof
Leaderboard	Sort all player/bot entries by net, wins, streak, or level
Players	Browse 99 bot profiles, records, cosmetics, country, and biography
Lobby	Inspect the combined Coin Toss/catalog waiting queue and live activity feed
Shop	Purchase and equip items from nine cosmetic categories
Season	Review VIP progress, rakeback, quests, login streak, achievements, and season timer
Wallet	Review balances, park/unpark coins, top up, referrals, transfer, limits, and session timer
Stats	Career totals, win rate, net result, global bias, z-score, jackpots
Verify	Recompute a stored Coin Toss/catalog proof from server seed, maker hash, taker hash, and game ID

4.1 Header controls

- Jackpot meter shows the current pool and pulses once armed.
- MAIN wallet chip shows the immediately usable main balance.
- Turbo offers 1×, 20×, 100×, 500×, and 1000× simulation rates.
- Admin link opens the companion panel; theme control switches dark/light mode.

5. Universal P2P matching, escrow, and settlement

5.1 Coin Toss queue flow

6. Player chooses a side and stake. Guard checks run before escrow.
7. Escrow atomically debits the allowed wallet segments and creates a waiting-room record.
8. A public bet is automatically accepted by a bot after a short delay; a private bet remains until manually taken or cancelled.
9. The bot takes the opposite side at the same stake and also escrows its stake.
10. The merged hash determines the result. The winner receives the post-fee pot in MAIN.
11. History, bot/player statistics, fees, jackpot contribution, quests, XP, feed, proof, and review flags are updated.

5.2 Catalog queue flow

12. Each catalog page collects the game-specific pick and stake.
13. The player may have one open bet per catalog game. Posting creates a kind=catalog queue entry and escrows the stake.
14. The page and global Lobby both display game, pick, stake, and waiting time.
15. A bot chooses a valid complementary/different pick and joins automatically. Reloaded queues are recovered by the background bot tick.
16. Before a bot joins, the player may cancel for a full refund to the original wallet segments.
17. After joining, the game-specific resolver reads proof bytes, applies split/carry rules, and settles balances and records.

5.3 Atomic wallet escrow

The non-main contribution cap is calculated first. The code then verifies that MAIN can fund the remainder before debiting any wallet segment. This prevents a failed stake from silently consuming BONUS, REFERRAL, or RAKEBACK balances.

Order	Segment	Rule
1	BONUS	Used first within the non-main cap
2	REFERRAL	Used after BONUS within the remaining cap
3	RAKEBACK	Used after REFERRAL within the remaining cap
4	MAIN	Funds the rest of the stake

5.4 Guard checks

- Maintenance mode must be off.
- Self-exclusion timer must have expired.

- Maximum 12 player bets per minute.
- Session loss limit must not be reached.
- Minimum stake: 10 coins.
- Maximum stake: $\min(1000, 100 + \text{player level} \times 100)$.
- Total wallet balance and required MAIN contribution must be sufficient.

5.5 Split and carry semantics

- Split: the available post-fee pot, including an existing carry pool, is divided between player and bot. Odd remainders go to the bot in the current implementation.
- Carry: no entrant is paid. The current post-fee pot is added to that game's persistent carry pool.
- Next non-carry result: the carry pool is added to the current post-fee pot and then reset to zero after payout or split.

6. Provably-fair RNG and verification

6.1 Coin Toss proof path

For each Coin Toss match, three random seeds are created in the browser. The hashes are combined as large integers and hashed again with the game ID.

Value	Formula / interpretation
makerHash	SHA256(makerSeed : gameId : stake : makerSide)
takerHash	SHA256(takerSeed : gameId : stake : takerSide)
commit	SHA256(serverSeed)
combined	$\text{BigInt}(\text{makerHash}) + \text{BigInt}(\text{takerHash}) + \text{BigInt}(\text{commit})$
finalHash	SHA256(combinedHex : gameId)
result	firstByte even $\rightarrow$ HEADS; odd $\rightarrow$ TAILS
jackpot	firstByte = 0x00 and pool armed

6.2 Catalog proof path

Catalog games use the same final combination, while each entrant hash also commits to the catalog game ID and pick: SHA256(seed : gameId : stake : catalogGame : pick). Individual resolvers read one or more bytes from finalHash.

6.3 Series and tournament RNG

- Series Cups derive repeated flip results by hashing the combined value with gameId and a flip nonce.
- Tournaments derive one flip per single-flip match or up to three flips per Bo3 match.
- Series/tournament player history stores outcomes and a verified marker, but not every standalone proof input. Coin Toss and catalog matches retain the full local verifier inputs.

6.4 Standalone verifier

18. Load the last retained Coin Toss or catalog proof, or paste the four required values.
19. The browser recomputes the server commitment, integer combination, final hash, first byte, result, and jackpot byte condition.
20. The result is local and does not call an external service.

Security limitation All seeds—including the value described as the server seed—are generated in the same browser process. This demonstrates the algorithm but is not a production commit/reveal service. A production system must publish commitments before acceptance, retain immutable timestamps, isolate server secrets, and settle on a server-authoritative ledger.

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7. Game rules and winner determination

7.1 Coin Toss

Player action	Bot match	Winner	Fee / special
Pick HEADS or TAILS and stake	Bot takes the opposite side	firstByte parity: even HEADS, odd TAILS	Regular pot fee; byte 00 may trigger jackpot when armed

7.2 Side-pick catalog games

Game	Allowed picks / bot match	How the winner is decided	Tie / carry
Over / Under	HIGH vs LOW	Byte 0 ≥ 128 is HIGH; otherwise LOW	None
Speed Round	HEADS vs TAILS	First five bytes become flips; majority side wins	Impossible with five flips
Tug of War	LEFT vs RIGHT	HEADS pulls left, TAILS pulls right; first position ± 3 wins. Resolver uses the proof-byte window and falls back to final direction if needed.	No formal tie
Even / Odd Sum	EVEN vs ODD	Bytes 0 and 1 are added; sum parity wins	None

7.3 Number games

Game	Pick / match rule	Winner calculation	Equal result
Closest Number	0–255; bot must choose a different number	Smallest absolute distance to byte 0	Split
Lucky Number Battle	0–255; different picks	Exact match is highlighted; otherwise closest to byte 0	Carry
Sum Prediction	0–510; different predictions	Closest prediction to byte 0 + byte 1	Split

Higher Byte	No pick	Player gets byte 0, bot gets byte 1; higher wins	Split
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7.4 Pattern games

Game	Pick	Resolution	Tie / carry
Pattern Race	One of eight 3-flip strings, e.g. HTH; bot chooses a different pattern	Read parity flips from proof bytes until either pattern appears consecutively	Carry if neither appears in the 32-byte proof window
Parlay Duel	One of eight 3-flip predictions	Generate three flips; most position-by-position matches wins	Carry on equal match count
Prediction Streak	One of eight configured 5-flip predictions	Generate five flips; most correct positions wins	Split on equal correct count
Blind Pick	HEADS or TAILS; bot may pick same or different	Different picks: side matching the flip wins. Same picks and matching flip: split. Same picks and nonmatching flip: carry.	Split/carry as described

7.5 Range games

Game	Zones	Resolution	Gap
Range War	0–63, 64–127, 128–191, or 192–255; bot takes a different zone	Byte 0 inside player zone → player; inside bot zone → bot	Any unclaimed quarter carries the pot
Bullseye	Same numeric quarters presented as target wedges	Byte 0 determines the winning wedge	Any unclaimed wedge carries the pot

7.6 Multi-flip drama games

Game	Pick / process	Winner	Tie / carry
Chain Reaction	Player and bot hold side calls; callers alternate for up to 10 proof bytes	First caller whose call differs from the flip loses	Both surviving 10 calls split
Elimination Ladder	HEADS vs TAILS	First selected side to appear three times reaches rung 3 and wins	None
Mirrored Coins	HEADS vs TAILS; player coin = byte 0, bot coin = byte 1	Different coin results: entrant whose own coin matches its pick wins	Matching coin results carry

7.7 Series Cups

Format	Target	Maximum flips	Winner
Bo3	First to 2	3	Entrant with target wins
Bo5	First to 3	5	Entrant with target wins
Bo7	First to 4	7	Entrant with target wins
Advantage Lead	by 2 after at least 3 flips	9	Higher score at a valid 2-point lead, otherwise higher score at cap

The first entrant is treated as the HEADS-side competitor and the second as TAILS-side. A Cup can be player-vs-bot or bot-vs-bot. The winner receives the two-entry pot minus Cup rake. Bot-only Cups never change the demo player's balance or statistics.

7.8 ×2 Double-or-Nothing room

- Available after a completed Coin Toss result.
- Challenge stake is twice the previous stake, limited by the player's current maximum stake.
- The challenge expires after ten minutes and may be cancelled for refund before acceptance.
- A demo bot normally accepts automatically; settlement reuses Coin Toss fairness and fee rules.

7.9 Tournaments

Format	Per-match rule	Bracket	Prizes
Single Flip	One merged-hash parity flip; first bracket entrant is HEADS side	Randomly shuffled 2/4/8/16 single-elimination bracket	After tournament rake: champion 75%, runner-up 25%
Bo3 Series	Up to three parity flips; first to two advances	Same single-elimination structure	After tournament rake: champion 75%, runner-up 25%

- Admin-created and automatic tournaments allow players and bots.
- The lobby preserves a short public join window; after roughly 4.5 seconds, bots fill every remaining seat.
- A full bot-only bracket starts and completes automatically without changing player statistics.
- Automatic tournaments use sizes 4/8/16, entries 50/100/250, and select Bo3 approximately 35% of the time.

8. Wallet, economy, jackpot, and fees

8.1 Wallet segments

Segment	Default Use
MAIN	1,000 Deposits, winnings, transfers, shop, majority of every stake
BONUS	250 Login, quests, levels, achievements, first top-up bonus; subject to non-main stake cap
REFERRAL	50 Referral rewards; subject to non-main cap
RAKEBACK	0 Claimed VIP rakeback; subject to non-main cap
BANK	0 Parked MAIN coins; not stakeable until unparked

8.2 Fee and rake model

Charge	Default	Base
Coin Toss / catalog pot fee	5%	Rounded percentage of the two-stake pot
Series Cup rake	5%	Rounded percentage of two Cup entries
Tournament rake	10%	Rounded percentage of all tournament entries
Transfer fee	2%	Rounded percentage of the sent amount; minimum effective fee in bot transfers is 1
Shop	Item price	100% recorded as shop revenue in the demo

8.3 Jackpot

- Default pool: 120 coins; default arming threshold: 50.
- A configured share of each regular/Catalog fee funds the pool; default funding rate is 10%, with a configured floor while retaining at least one fee coin for the house calculation.
- Coin Toss jackpot trigger: final first byte equals 0x00 and pool is armed (theoretical 1/256).
- Default payout: 50% of the current pool. The remainder rolls over.
- Catalog matches fund the pool but do not currently execute a jackpot trigger; the active trigger is in Coin Toss and legacy internal game paths.

8.4 Economy accounting

- House revenue buckets: regular fees, Cup rake, tournament rake, shop sales, and transfer fees.
- Costs: promotion cost and admin comps.
- Taps count created play coins; sinks count removed coins. The admin panel displays sink/tap ratio and a health verdict.
- Demo bot auto-top-ups maintain liquidity and are explicitly not counted as house revenue.

9. Progression, VIP, quests, achievements, and seasons

9.1 XP and levels

Level = floor(sqrt(XP / 60)) + 1. Stakes increase XP. On level-up, the configured reward goes to BONUS; if a level is not listed, fallback reward = level × 50. The default table configures levels 2–25 from 100 to 1,250 coins.

9.2 VIP tiers

Tier	Monthly wager threshold	Rakeback	Shop discount
Starter	0	0%	0%
Silver	1,000	4%	0%
Gold	3,000	6%	0%
Platinum	8,000	8%	0%
Diamond	20,000	12%	0%
Black Diamond	50,000	15%	10%
Royal	75,000	17%	20%
Legend	100,000	20%	30%

Coin Toss accrues rakeback as tier percentage × half the match fee. Claiming moves accrued rakeback into the RAKEBACK wallet. VIP thresholds and rates are editable in admin, and the monthly counter can be reset manually.

9.3 Quests

Quest	Target	BONUS reward
Settle games	3	50
Win a game	1	75
Play a Series Cup 1		50

Current limitation The UI labels these as daily quests, but the present build does not implement a scheduled UTC quest reset. Claimed state persists in localStorage until reset or state replacement.

9.4 Login streak

- Uses the browser's local calendar day.
- Reward = 50 + 25 × (day − 1), capped at 175 per day, paid to BONUS.
- Missing the preceding local day resets the streak to day 1.

9.5 Achievements

Achievement	Condition	Reward
First Flip	Settle first game	30
On Fire	3-win streak	75
Unstoppable	5-win streak	150
Jackpot!	Win jackpot	300
Cup Winner	Win Series Cup	100
Champion	Win tournament	300
Big Baller	Stake 500+	100

Night Owl	Play 10 games	50
Frequent Flipper	Play 25 games	100
Double Digits	Win 10 games	100
High Roller	Reach level 5	100
Comeback Kid	Win after at least two consecutive losses	75
Century Club	Settle 100 games	250
Veteran	Settle 250 games	500
Quarter Century	Win 25 games	250
Winning Machine	Win 100 games	750
Volcanic	Reach a 10-win streak	500
Cup Specialist	Win 5 Series Cups	300
Bracket Boss	Win 3 tournaments	500
Game Explorer	Play 5 catalog games	150
Catalog Master	Play all 17 catalog games	750
Collector	Own 5 paid cosmetics	150
Super Collector	Own 15 paid cosmetics	400
Silver Status	Reach Silver VIP	100
Gold Status	Reach Gold VIP	250
Profit Maker	Reach +1,000 career net	300
Double-Digit Level	Reach level 10	300
Elite Level	Reach level 20	750
Generous	Complete 5 player transfers	150

9.6 Seasons

The default season ends at 20:00 UTC on the last day of the current month. The UI presents a 500-coin champion reward and 250-coin runner-up reward, but automated season-end ranking and payout execution are not implemented in the current JavaScript.

10. Cosmetics shop and social features

10.1 Shop categories

Category	Count	Examples / behaviour
Coin skins	15	Classic, silver, neon, galaxy, rainbow, Legend Prism; changes coin rendering
Flags	12	Country or Earth flag beside player identity
Personas / avatars	12	Hero, ninja, robot, royal, dragon, etc.
Frames	8	Thin, gold, fire, ice, neon, rainbow effects
Chat colours	8	Colour metadata for identity/taunt styling
Victory FX	8	Confetti, coins, fireworks, money, phoenix, dragon
Themes	8	Midnight, ocean, forest, sunset, space, casino, aurora, cyber
Sound packs	6	Standard, 8-bit, orchestral, DJ, trailer, choir
Premium emojis	15	Additional reaction choices

- Purchases debit MAIN and are recorded as shop revenue/sinks.
- Purchased non-emoji items are equipped immediately; owned items can be re-equipped later.
- Some items require a VIP tier; discounts apply at Black Diamond, Royal, and Legend.

10.2 Social and presentation features

- Optional 24-character taunt on Coin Toss posts.
- Reactions on recent games: three free reactions plus owned premium emojis; five-second player cooldown.
- Bot profiles show country, biography, record, balance, cosmetics, streak, and jackpot count.
- Activity feed records match results, jackpots, Cups, tournaments, purchases, transfers, progression, and top-ups.
- Leaderboard supports net profit, wins, streak, and level sorting.

11. Bots, liquidity, transfers, and Turbo

11.1 Bot roster

The default roster contains 99 named bots with avatar, flag, balance, level, country, title, biography, skin, and runtime statistics. The first 30 are explicitly authored; the remainder are procedurally generated at page load and migrated into older state.

11.2 Bot behaviours

Behaviour	Current implementation
Coin Toss liquidity	Seeds approximately 12 waiting bets and accepts public player bets
Catalog liquidity	Joins every catalog waiting-room type with a valid complementary/different pick; also plays bot-vs-bot catalog matches
Series	Opens public Cups, joins player Cups, and completes bot-only Cups
Tournaments	Fills all remaining seats after the join window and completes bot-only brackets
Shopping	Occasionally buys affordable skins, avatars, or frames; may equip purchased skins Occasionally sends 50/100/250/500 coins to another bot, pays transfer fee,

Transfers	updates both balances and admin history
Auto top-up	Triggers below 100 coins; adds a bot-specific 400–1,600 base amount. The first bot top-up receives +50% when the promo is enabled.

11.3 Turbo simulation

Setting Bot Coin Toss workload per background tick Purpose

1×	6 flips	Normal live ecosystem
20×	20 flips	Light simulation
100×	100 flips	Stress test
500×	500 flips	High-volume economy/RNG test
1000×	1,000 flips	Maximum built-in stress test

The background tick runs approximately every 2.2 seconds and is protected against overlap. At high volume the loop yields every 100 flips to reduce UI starvation. Feed messages are suppressed aggressively, and persistence is throttled to four-second intervals.

Performance warning 1000× intentionally performs thousands of SHA-256 operations per cycle and can consume substantial CPU, battery, memory, and localStorage bandwidth. Use it for short stress tests, not normal play.

12. Tournament and Series operations

12.1 Series lifecycle

21. Player or bot creates a Cup with format and entry; first entry is escrowed.
22. A player may join a bot Cup before another bot does. Player Cups are normally filled by bots.
23. Once two entrants exist, the Cup leaves the open list and settles its flip series.
24. Cup rake is recorded; winner receives the post-rake pot; bot/player stats are updated appropriately.
25. Bot-only Cups generate activity and economy records without touching the demo player.

12.2 Tournament lifecycle

26. Admin or automatic scheduler creates an open bracket with size, entry, rake, and single/Bo3 format.
27. Bots initially fill all but one slot to provide a short player join opportunity.
28. After roughly 4.5 seconds, bots fill every remaining slot.
29. Entrants are shuffled; rounds play until champion and runner-up are known.
30. Rake is removed; the remaining pool pays 75% to champion and 25% to runner-up.
31. The completed tournament is compacted before persistence to avoid duplicating full bot objects.

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13. Admin panel

Tab	Capabilities
Dashboard	Revenue tiles, house P&L, jackpot, feed, RNG chart, z-score, jackpot hit rate
Rates & Jackpot	Regular/Cup/tournament/transfer rates, jackpot funding/floor/arm/payout, stake cap, transfer limits
VIP & Levels	Edit eight VIP tiers/rakeback, reset table, end month, edit/add level rewards
Tournaments	Create 2/4/8/16 brackets, choose single or Bo3, set entry and rake, monitor status/champion
Promotions	Pause first top-up +50% promo, create/pause/delete campaign metadata, set broadcast
Economy	Taps/sinks, ratio, health verdict, revenue simulator, revenue-funded comp tool
Ops & Players	Maintenance/features, demo player summary, review flags, queue analytics, open P2P queue, bot transfer history
Audit & Export	Admin audit trail, ledger/game/full-state exports, reset, player history, all catalog results

13.1 Configurable rates and rules

- Regular bet fee: 0–25%.
- Series Cup rake: 0–25%.
- Tournament rake: 0–25%.
- Transfer fee: 0–10%.
- Jackpot funding: 0–50%; floor 0–10; arm 0–1,000; payout 1–100%.
- Non-main stake cap constrained to 10–20%; transfer minimum and daily cap editable.

13.2 Feature toggles

- Maintenance mode blocks new betting while allowing in-flight settlement.
- Auto-match queue, demo bots, quests, and daily login are persisted feature flags.
- First top-up +50% promo can be paused/resumed independently in Promotions.
- A ×2 display toggle exists in the current admin UI, but it is not wired to a persisted feature flag; ×2 remains available in the player build.

13.3 Monitoring and review

- Queue monitor displays Coin Toss and catalog game name, player, pick, stake, and wait time.
- Catalog results include player/bot names, both picks, stake, fee, result, detailed resolution, and proof hash prefix.

- Bot transfer monitor displays time, sender, receiver, net sent, and fee.
- Payouts of at least 400 coins and jackpots create review flags.
- RNG health reports HEADS/TAILS counts and z-score; only actual HEADS/TAILS result rows are plotted.

14. Promotions and top-ups

14.1 First top-up promotion

- Default offer: +50% of the top-up amount to BONUS on the first eligible top-up.
- Minimum top-up input enforced by the handler: 100 coins.
- Admin can pause the offer. While paused, regular player top-ups add MAIN only. The same toggle controls the +50% bonus on each bot's first automatic top-up.
- After the offer is claimed, future top-ups add MAIN only.
- Bonus cost is recorded as promotion cost and created coins as taps.

14.2 Campaign manager

Admin can create scheduled campaign metadata of type deposit bonus, flat credit, or cash drop; pause/resume or delete it; and inspect live/scheduled/ended state. Broadcast text appears globally in the player application.

Current limitation Generic campaign redemption is not wired to a player claim workflow. The campaign manager currently stores and displays campaign metadata; the first top-up +50% mechanic is the only implemented top-up promotion payout path.

14.3 Broadcast and maintenance messaging

- Broadcast is limited to 120 characters and appears above every player screen.
- Maintenance banner clearly states that new betting is paused while in-flight games can settle.

15. Responsible gaming and controls

Control	Behaviour
Session loss limit	Blocks new bets once session net is at or below the negative limit
Self-exclusion	Demo 60-second betting lock stored in memory for the current page session
Reality check	Session timer; recurring message every five minutes
Auto-rebet stop	Automatically disables auto-rebet at session net ≤ -200
Bank / vault	Moves MAIN outside the stakeable wallet until manually unparked
Rate limit	Maximum 12 player bets per rolling minute
Play-coins positioning UI and footer state that the product uses play coins only	

Production requirement Real responsible-gaming controls must be server-enforced, durable across devices/sessions, protected from administrator override, and accompanied by jurisdiction-appropriate limits, cooling-off, exclusion, and support workflows.

16. Persistence, audit, exports, and synchronization

16.1 Persistence

- Primary state key: `tossmatch_v8`.
- Per-tab heartbeat keys identify page instances but are not a complete concurrent-write lock.
- Game history capped at 60; player ledger capped at 200; feed capped at 60; admin audit capped at 50; catalog log and bot transfer history capped at 100.
- Clearing browser site data removes all progress and configuration.

16.2 Audit and review

- Admin rate, jackpot, rule, VIP, promotion, broadcast, comp, tournament, exclusion, and reset actions are timestamped where handlers call `audit()`.
- Player top-ups and selected player actions create audit entries.
- The audit trail is a mutable local array; it is not tamper-evident.

16.3 Exports

Export	Contents
Ledger CSV	Timestamp, transaction type, delta, note, and MAIN balance for recorded player-ledger events
Games JSON	Player game-history array including available proofs and result details
Full state JSON	Entire local demo state including config, wallets, bots, histories, queues, and economy

17. Default configuration reference

Parameter	Default
Regular fee	5%
Cup rake	5%
Tournament rake	10%

Jackpot funding	10% of regular/catalog fee
Jackpot floor	1
Jackpot arming threshold	50
Jackpot payout	50%
Starting jackpot	120
Non-main stake cap	20%
Transfer fee / min / cap	2% / 10 / 500 per day
Starting player balances	MAIN 1,000; BONUS 250; REFERRAL 50
House capital	100,000
Bot count	99
First top-up promo	Enabled, +50% to BONUS
Turbo default	1× in state; UI synchronized on boot

18. End-to-end operational flows

18.1 Player posts a catalog match

32. Open Games and select a catalog title.
33. Choose a valid side, number, pattern, or zone; Higher Byte requires no pick.
34. Enter stake and press “Post to waiting room”.
35. Confirm the bet appears on the game page, Lobby, and admin open queue.
36. Allow a bot to join. Confirm queue removal and resolution.
37. Inspect result details, picks, delta, fee, carry state, and proof.
38. Open admin Catalog Results and verify the same match is listed.

18.2 Admin pauses first top-up promotion

39. Open Admin → Promotions.
40. Switch “First top-up +50% promo” off.
41. Open Wallet in the player app; confirm the paused message.
42. Top up at least 100 coins; verify MAIN increases and BONUS does not.
43. Re-enable the toggle; if never claimed, the first-promo eligibility remains available.

18.3 Bot-only tournament

44. Create a tournament in admin, selecting single flip or Bo3.
45. Observe bots fill all but one seat during the public window.
46. Do not join. After the timeout, verify the final bot joins and bracket runs.
47. Confirm champion/runner-up, tournament rake, bot balances/stats, activity feed, and unchanged player net.

18.4 Stress test

48. Export or reset state as needed before testing.
49. Select a Turbo level; use 1000× only for short runs.
50. Watch global game count, jackpot pool, bot records, house fees, sink/tap ratio, z-score, shop activity, and transfers.
51. Return Turbo to 1× and allow the four-second persistence interval to save.

19. Failure handling and edge cases

Condition	Current behaviour
Insufficient MAIN under non-main cap	Escrow rejects before any segment is debited
Cancel unmatched Coin/Catalog bet	Original segment split is refunded
Catalog bot unavailable	Bet remains in waiting room; later background tick retries
Busy engine	Background tick skips; catalog bot join schedules a retry
Match error after bot joins	Player split is refunded and bot stake restored
Expired >2	Challenge is filtered/refunded by room logic when processed
Tournament lacks bots with entry balance	Lobby stays open until sufficient bots are available/auto-topped up
localStorage write failure	Save exception is swallowed; in-memory session may continue
Multiple tabs write concurrently	Storage events reload state, but no server-grade transaction/locking exists
Carry game result	Post-fee pot persists by catalog game ID until a non-carry outcome

20. Testing and acceptance checklist

20.1 Functional smoke checks

- ☐ Player and admin HTML parse with no JavaScript syntax errors.
- ☐ Player boot creates state, 99 bots, 17 catalog games, queue, and navigation.
- ☐ Each catalog game accepts a pick, creates an escrowed queue entry, auto-matches a bot, and writes catalogLog.
- ☐ Catalog cancellation refunds before matching.
- ☐ Coin Toss result, proof, fee, jackpot contribution, and history are updated.
- ☐ Bot-only Series and Bo3 tournament complete without changing player net.
- ☐ Paused first top-up promo adds MAIN only.
- ☐ Bot transfer changes sender/recipient balances and records fee/history.
- ☐ Admin renders open queue, catalog history, game details, bot transfers, and promo toggle.
- ☐ 1000× control is present and simulation loop yields periodically.

20.2 Recommended regression matrix

Area	Cases
Wallet	Each source segment, insufficient MAIN, park/unpark, transfer limits, promo on/off
Matching	Public/private Coin, catalog post/cancel/reload/retry, busy-engine retry
Games	Every win/lose/split/carry branch; boundary bytes 0, 127, 128, 255
Series	Bo3/Bo5/Bo7/Advantage; player-bot and bot-bot
Tournaments	2/4/8/16, single/Bo3, player champion/runner/eliminated, bot-only
Fairness	Recompute stored Coin and catalog proofs; malformed verifier input
Admin	Every rate/toggle, queue sync, exports, reset, revenue calculations
Performance	1× through 1000×, localStorage quota, long-running bot economy

21. Production-readiness gaps

Summary The current build is a strong interactive prototype and simulation harness, not a production wagering platform.

Gap	Why it matters	Production direction
No backend authority	Clients can modify balances, RNG, admin state, and history	Server-side accounts, transactional ledger, signed events, authorization
Local “server seed”	Commit/reveal is not independently controlled	Server commitment service, published pre-match commitment, auditable reveal
No real multiplayer transport	All participants are local bots	Authenticated realtime matchmaking, durable queues, idempotent settlement
No secure admin access	admin.html is directly accessible	RBAC, MFA, approval workflows, immutable audit
No payment/compliance stack	Play coins only	Licensed payments/custody, KYC/AML, geofencing, sanctions, limits
Mutable local audit	History can be edited or reset	Append-only event store, retention, reconciliation, monitoring
Incomplete promo/season automation	Campaign metadata and season display exceed payout automation	Rules engine, eligibility, claims, expiry, scheduled jobs
Client-only RG state	Limits can be bypassed by reload/device change	Server-enforced durable RG controls
Math not certified	Some catalog mechanics use bounded proof windows/demo carry logic	Formal probability review, RTP/rake validation, independent certification
Concurrency risk	Multiple tabs may overwrite state	Versioned transactions, optimistic locking, idempotency keys

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Appendix A — Glossary

Term	Meaning
Armed jackpot	Pool has reached the configured threshold and can pay on Coin Toss byte 00
Bot liquidity	Synthetic participants that post, join, and settle matches in the demo
Carry pool	Post-fee pot retained for the next non-carry result of the same catalog game
Commitment	SHA-256 hash of the server-seed value before reveal in the fairness model
Comp	Admin goodwill credit funded from available net house revenue
Escrow	Stake removed from available balances and attached to an open match record
MAIN	Primary balance used for most stake funding, purchases, transfers, and winnings
Merged hash	Final SHA-256 input built from both entrant hashes and the server commitment
Non-main cap	Maximum stake percentage funded by BONUS, REFERRAL, and RAKEBACK combined
Rake	Amount removed from a Cup or tournament pool before prizes
Rakeback	VIP reward accrued from Coin Toss fee contribution
Sink	Coins removed from the player/bot economy through fees, rake, transfers, or shop
Tap	Coins created through top-ups, rewards, promotions, comps, or bonuses
Turbo	Multiplier controlling bot Coin Toss simulation volume per background tick
z-score	Standardized deviation of HEADS count from a fair 50/50 expectation

Appendix B — Quick rule index

Mode	Primary proof bytes / loop	Decisive condition
Coin Toss	firstByte	Even HEADS / odd TAILS; 00 jackpot if armed
Over/Under	byte 0	≥128 HIGH
Speed Round	bytes 0–4	Majority parity
Tug of War	successive bytes	Random walk reaches ±3
Even/Odd Sum	bytes 0–1	Sum parity
Closest / Lucky Battle	byte 0	Smallest distance
Sum Prediction	bytes 0–1	Closest to sum
Higher Byte	bytes 0–1	Higher value
Pattern Race	up to 32 parity bytes	First selected pattern
Parlay Duel	bytes 0–2	Most matching positions
Prediction Streak	bytes 0–4	Most matching positions
Blind Pick	byte 0	Secret-pick branch rules
Range / Bullseye	byte 0	Result inside claimed zone
Chain Reaction	up to 10 bytes	First caller miss
Elimination Ladder	successive bytes	First side to three appearances
Mirrored Coins	bytes 0–1	Different results decide; same carries
Series	nonce hashes	First to target / Advantage lead
Tournament	per-match nonce hash(es)	Advance until champion

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Appendix C — v9.6 implementation update

This appendix records changes made after the original documentation build. It is part of the authoritative project reference.

Change	Current behaviour
Default balance	New/reset demo accounts start with MAIN 1,000 instead of 5,000. An inactive v8 state with zero games and untouched MAIN 5,000 is migrated to 1,000; active accounts are preserved.
Bot top-ups	Automatic bot top-up now triggers below 100. Base top-up varies from 400 to 1,600 and cannot repeat consecutively for the same bot. The first bot top-up receives +50% when the admin promo is enabled. Top-ups record base, bonus, reason, count, and total.
Catalog earnings	Catalog fees are separated from Coin Toss fees in house accounting. Admin Dashboard/P&L and Catalog Results display catalog earnings, turnover, match count, and open carry pools.
Admin data controls	Queue, tournaments, review flags, transfers, bot top-ups, audit, player games, and catalog results use a 20-row page size with relevant filters, sorting, previous/next controls, and row counts.
Bot activity	Normal Coin Toss bot workload increases from 3 to 6 per tick. Catalog activity scales from 2 matches per normal tick up to 12 at high Turbo. Transfer, shopping, Cup, and tournament activity probabilities are increased.
Levels	Default configured level rewards now extend through level 25, from 100 at level 2 to 1,250 at level 25. Older saved configurations merge in the new defaults.
Achievements	Seventeen achievements were added for long-term games/wins/streaks, Cups, tournaments, catalog exploration, cosmetics, VIP, profit, levels, and transfers.

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Appendix D — Catalog manual-take and activity update

Catalog matchmaking now supports automatic matching and player-initiated manual acceptance, mirroring Coin Toss, Series Cups, and tournament lobbies.

Change	Current behaviour
Bot-posted bets	Bots maintain open catalog bets with game, pick, stake, owner, and wait time on the game page, Lobby, and admin queue.
Manual take	Players can take bot bets. Two-option games auto-select the complement; number, pattern, and zone picks must differ. Blind Pick may use the same pick.
Auto-match	Player posts still receive automatic bot joins. Bot posts remain manually takeable, then may auto-match another bot after a delay.
Escrow	The posting bot is debited at post time; the manual taker is checked and escrowed at acceptance.
Activity	Normal catalog activity is six direct bot matches per tick, aligned with six normal Coin Toss matches; Turbo scales catalog activity up to 25.
Inventory	The simulator progressively maintains bot entries across all 17 catalog game waiting rooms.

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Appendix E — v9.6 VIP implementation

The VIP implementation follows the document’s explicit eight-row complete tier table: Starter, Silver, Gold, Platinum, Diamond, Black Diamond, Royal, and Legend. The conflicting six-tier and incomplete ten-tier references were not used.

Tier	Monthly wagered Rakeback Implemented benefits		
Starter	0+	0%	Basic access; quests; jackpot eligibility
Silver	1,000+	4%	Blue chat colour; 1 free premium emoji
Gold	3,000+	6%	Gold avatar frame; priority queue; 2 free premium emojis
Platinum	8,000+	8%	Permanent Platinum VIP coin skin; 2 free emojis
Diamond	20,000+	12%	Permanent Diamond VIP skin; 5% tournament discount; 3 free emojis
Black Diamond	50,000+	15%	Permanent animated frame; birthday eligibility; tournament −10%; shop −10%; 5 emojis
Royal	75,000+	17%	Royal Crown title; tournament −10%; shop −20%; priority support entitlement; 8 emojis
Legend	100,000+	20%	Legend Prism and Eternal Flame skins; tournament −15%; shop −30%; all emojis; gold name; early-access entitlement

Operational rules: tier is based on current UTC-month wagered volume; the counter auto-resets when the UTC month changes or when admin ends the VIP month. Pending rakeback and permanent cosmetic/emoji unlocks survive. Discounts, queue priority, titles, support, and early-access status require the current monthly tier. Coin Toss and player catalog matches accrue $\text{round}((\text{fee} / 2) \times \text{rakeback}\%)$. Admin can edit thresholds and rates within the existing 0–20% controls; fixed tier benefit bundles follow tier number.

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Appendix F — v9.6 B1–B4 feature implementation

All 23 feature ideas listed under document sections B1 through B4 are represented by dedicated player screens and working local-demo interactions. The main navigation now includes Community, New Games, Progress+, and Economy+.

Section	Feature	Implemented demo behaviour
B1 Social	S1 Friends	Add/remove online bot friends, direct challenges, friend-aware controls
B1 Social	S2 Private Rooms	Invite-code/friends-only rooms with custom game and stake range
B1 Social	S3 Profiles	Public player statistics, cosmetics, achievements, and recent games
B1 Social	S4 Spectator	Watch recent live bot/player catalog matches with pot, picks, result, and proof
B1 Social	S5 Gifting	Gift MAIN coins or owned paid cosmetics to friends
B1 Social	S6 Chat	Lobby chat with bot replies, premium emoji support, mute and block
B1 Social	S7 Clans	Create clan, add friends, team score, team matches, clan leaderboard
B2 Games	G1 Lucky Wheel	Daily free spin plus 50-coin paid spins with coin/cosmetic prizes
B2 Games	G2 Scratch Cards	Common/Rare/Epic paid cards; reveal 9 cells; 3+ matches pay
B2 Games	G3 Dice Roll	Exact double at 30× or low/high sum at 1.8×
B2 Games	G4 Raffle	10-coin tickets, bot pool, immediate demo draw, 80/20 winner/house split
B2 Games	G5 Multiplier Ladder	P2P player/bot ladder with rung or bust result
B2 Games	G6 War Card	P2P card ranks 2–Ace with automatic war on ties
B3 Progress	P1 Battle Pass	Monthly free/premium milestones and claims
B3 Progress	P2 Login Calendar	Seven visual days; escalating BONUS; day-7 cosmetic and 250
B3 Progress	P3 Weekly Challenges	Wins, distinct games, and streak challenges with larger rewards
B3 Progress	P4 Prestige	Level 10 reset, badge/rank, Rainbow frame, permanent +5% XP per rank
B3 Progress	P5 Skill Matchmaking	Rookie/Bronze/Silver/Gold/Elite tiers and preferred same-tier bot matching
B4 Economy	E1 Mystery Crates	Four crate tiers, 1–3 random cosmetics, rarity floor
B4 Economy	E2 Trading Post	Bot listings, player listings, simulated sales, 10% house fee
B4 Economy	E3 Staking	Vault deposits/withdrawals, 1% completed-week interest, claim cap
B4 Economy	E4 Subscriptions	Plus/Pro/Elite 30-day tiers, cosmetics, themes, 2× quest perks
B4 Economy	E5 Boosters	2× XP for one hour and +5% rakeback for 24 hours

Fast bot matching: normal direct activity is six Coin Toss and six catalog matches per background cycle. Bots also maintain waiting bets. Bot Coin Toss entries auto-match from wait age 1 in batches of four; catalog entries auto-match from age 1 in batches of eight. Player-posted catalog bets are joined without a probabilistic delay. Series bot joins use a short 100–280 ms delay.

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Appendix G — v9.6 premium UI and admin upgrade

The player and admin applications now share a premium dark visual system with responsive sidebar navigation, glass-gradient surfaces, stronger hierarchy, compact data presentation, and mobile horizontal navigation.

Area	Upgrade
Player Home	New default dashboard with bankroll, career net, jackpot, level, live opportunities, VIP journey, activity pulse, and feature shortcuts.
Player Navigation	Desktop vertical menu grouped into Overview, Play, Community, Progress & Economy, and Account. Mobile/tablet returns to horizontally scrollable navigation.
Player Header	Profile/VIP chip, refined wallet and jackpot cards, premium backgrounds, shadows, hover states, and responsive controls.
Admin Command Center	Hero status, live local clock, health alerts, quick actions, premium KPI cards, revenue, feed, jackpot, and RNG views.
Admin Sidebar	Grouped Command, Commercial, Engagement, and Governance navigation with responsive mobile fallback.
Admin Feature Hub	Dedicated B1–B4 telemetry for Community, Arcade+, Progress+, and Economy+, plus operational controls.
Data Inspector	Clickable player game and catalog result rows open a slide-in record drawer with match, financial, resolution, and proof details.
Quick Actions	Maintenance toggle, jackpot seed, tournament creation, broadcast composer, and state export from the dashboard.
Data UX	Sticky table headers, row hover states, upgraded filters, pagers, inputs, alerts, and responsive cards.

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Appendix H — v9.6 new-feature UI and description correction

Every B1–B4 feature sub-screen now uses a consistent premium feature header containing its source code, feature name, corrected

operational description, effort level, revenue/retention angle, live-demo status and three concise “how it works” points.

Area	Correction
B1 corrections	Friend bets sort first and show a FRIEND badge; profiles include a win-rate graph; spectator view includes an animated stage; coin gifting enforces minimum/daily cap/fee; chat exposes owned emoji buttons; clans include team matches and clan tournaments.
B2 corrections	Each game description now states cost, match or draw rule, payout rule, tie behaviour and whether it is P2P. Scratch, Ladder and War activity is persisted for admin telemetry.
B3 corrections	Descriptions distinguish monthly Battle Pass resets, automatic calendar claims, weekly isolation, Prestige reset effects and skill-match fallback behaviour.
B4 corrections	Descriptions clarify rarity floors, 10% Trading Post fee, completed-week staking interest and cap, 30-day subscription benefits, booster durations and effective rakeback cap.
UI consistency	All 23 features use feature-intro banners, status/revenue pills, three-point explanation cards, refined hover states and responsive mobile layouts.
Admin consistency	Feature Hub cards now include corrected category descriptions for Community, Arcade+, Progress+ and Economy+ monitoring.

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Appendix I — v9.6 mobile UI correction

The player and admin interfaces were corrected for narrow mobile browsers after reviewing the supplied iPhone screenshot.

Area	Correction
Navigation	Player and admin menus are forced into one non-wrapping horizontal row below 1,120 px, with touch momentum scrolling and hidden scrollbars. Feature, game, shop and chip sub-navigation use the same pattern.
Forms	All text, number, select and textarea controls now share dark-theme styling, full-width sizing, visible focus states, minimum-width protection and iOS-safe appearance. Labels are block-level and no longer collide with adjacent fields.
Private Rooms	Room name, game, stake limits and access controls collapse into a single-column mobile form with a full-width action button.
Header	At 800 px and below, the player header becomes a compact two-row grid; secondary profile, Turbo and admin-link controls are hidden. Wallet and jackpot remain visible.
Content grids	Cards, two/three/four-column grids, feature panels, profile statistics and action rows collapse cleanly without horizontal overflow.
Feature sub-screens	Feature tabs use horizontal touch scrolling; intro cards, explanation points, calendars, scratch cells and game actions reflow for mobile.
Admin	Filters and action rows stack, tables scroll horizontally, quick actions collapse, VIP editor rows reflow and the record drawer becomes full-width.

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Appendix J — v9.6 player history and advanced statistics

A dedicated Player History screen and expanded Statistics screen now preserve and expose activity that was previously available only as transient UI state.

Area	Implemented history/statistics
All Games	Up to 200 player game records with opponent, stake, fee, result detail, net delta, payout and proof when available.
P2P Catalog	Player-specific catalog history, plus a recent per-game history section directly on each catalog game page.
Friend Challenges	Opponent, sides, result, payout, delta and game ID are retained.
Private Rooms	Room creation/deletion/start lifecycle, invite code and settings, details modal, recent room games, and Coin Toss or selected catalog room results.
Clan Games	Team match and four-clan tournament result, opponent/placement, score impact and BONUS reward.
Arcade+	Lucky Wheel, Scratch Cards, Dice, Raffle, Multiplier Ladder and War results with cost, payout and net amount.
Progress+	Premium pass purchases, pass/weekly claims and Prestige events.
Economy+	Crates, Trading Post purchases/sales, staking movements/interest, subscriptions and boosters.
Social	Friend changes, gifting and sent chat activity.
History tools	Category tabs, search, newest/oldest/highest-amount sorting, 20-row pagination, detail modal and JSON export.
Advanced stats	Lifetime wagered, total payouts, maximum single payout, fees paid, average stake, average payout, return on wagered, draws/carries and game-family participation.
Payout trend	Recent 30-game visual chart with positive/negative bars and value tooltips.
Migration	Older saved games seed the new aggregate financial fields when the metrics version is first introduced.

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Appendix K — v9.6 feature consistency and before/after register

Version 9.5 synchronizes the player UI, admin UI, project documentation, feature register, changelog and audit report. The following matrix is the authoritative high-level before/after summary.

Area	Before	Current v9.6
P2P matching	Delayed bot acceptance	Manual take, player auto-match, bot-posted bets and fast bot-vs-bot settlement
VIP	Thresholds and rakeback focus	Complete eight-tier rewards, discounts, priority, titles and monthly reset
B1–B4	Twenty-three ideas in source document	Twenty-three dedicated working local-demo features
History	Recent games only; several transient feature results	Nine searchable history categories, details, pagination and export
Statistics	Games, wins, net and streak	Lifetime wager, total/max payout, fees, averages, ROI, breakdown and trend
Mobile UI	Wrapped navigation and overlapping native controls	Touch-scroll navigation, themed controls and responsive one-column forms
Admin	Top-tab operational pages	Command Center, Feature Hub, alerts, quick actions, paging and record drawer
Audit accounting	Double-counted sink ratio and incomplete paths	Corrected sink ratio, Catalog gross, Raffle, direct payout and Trading Post accounting

The complete feature-by-feature status, production boundary and recommendations are maintained in TossMatch_Feature_Register.md. Release chronology is maintained in CHANGELOG.md. Audit findings and corrective actions are maintained in TossMatch_Audit_Report.md.

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Appendix L — v9.6 Auto Bet and Feature Directory

Version 9.6 adds a configurable Auto Bet stop and a complete category-wise Admin Feature Directory, then synchronizes the change across player UI, admin UI, documentation, feature register, changelog and audit report.

Area	Before	Current v9.6
Auto Bet default	Hard-coded stop at session net −200	−200 remains default; player may select −100, −200, −500 or enter −50 to −10,000 in steps of 50.
Auto Bet UI	Toggle only	Premium configuration panel shows current stop, active state, presets and custom field.
Auto Bet settlement	Compared session net with constant −200	Settlement compares against settings.autoRebetStop and reports the configured stop in the message.
Feature discovery	Feature details distributed across screens and documents	Admin Feature Directory groups all features by category with search, status filters and quick links.
Feature statuses	Implemented features were listed separately from audit recommendations	Directory labels Implemented, Partial/demo-only and Suggested next.
Quick links	Manual navigation required	Directory opens the correct player tab/sub-feature through URL deep links or switches admin tabs directly.
Additional suggestions	B1–B4 implemented	B5 Platform, B6 Security and B7 Responsible Gaming items are included as implemented/partial/suggested roadmap entries.

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Appendix M — v9.6 Suggested Catalog and Arcade roadmap

Status boundary: all CAT18–CAT29 and G7–G16 records are Suggested roadmap concepts only. They are visible in the Admin Feature Directory and summarized in Player What’s New, but they have no playable player screen, settlement handler, wager path, or claim flow in v9.6.

M.1 Shared Catalog operating requirements

Any roadmap concept promoted into the playable P2P Catalog must preserve the common model established by the supplied 2-player game catalog:

- Exactly two entrants per match; player-vs-player semantics with bots permitted as local-demo stand-ins. No Catalog concept is a vs-house game.
- A per-game waiting room, posted-stake escrow, automatic matching, manually taking an available compatible bet, automatic bot joining without excessive wait, bot postings, and bot-vs-bot liquidity.
- Published compatibility rules for opposite picks, different picks, non-overlapping zones, or no-pick games; incompatible entries remain waiting rather than being forced together.
- Published winner, tie/split, carry, timeout, cancellation, fee and rounding rules before stake confirmation.
- Provably fair result material in the match record, plus player per-game history and Admin result/detail visibility.
- Catalog fee/earnings attribution in Admin revenue and P&L reporting without double-counting recorded sinks.
- Server-authoritative escrow and settlement, authenticated realtime matchmaking, abuse detection, tested deterministic rule functions, accessibility checks, and legal/regulatory review before production.

M.2 Suggested Catalog concepts

Code / game	Proposed winner determination	Waiting-room compatibility	v9.6 status
CAT18 — Rock Paper Scissors Duel	Hidden simultaneous ROCK/PAPER/SCISSORS choices. Standard dominance decides; identical choices split.	Distinct hidden choice; all pairs are compatible.	Suggested — not playable
CAT19 — Closest to 21	Proof-derived cards are dealt to both entrants. Closest total to 21 without exceeding wins; define both-bust and equal-total splits.	No pick or distinct strategy choice; two seats.	Suggested — not playable
CAT20 — Triple Coin Majority	Entrants take opposite HEADS/TAILS sides. Majority of	Opposite sides only.	Suggested — not playable

	three proof-derived flips wins.		
CAT21 — Sequence Builder	Entrants choose different two-symbol starters. First starter to complete the published target pattern in a proof sequence wins.	Different valid starters.	Suggested — not playable
CAT22 — Dice Sum Duel	Two proof-derived dice per entrant. Higher sum wins; equal sums split.	No pick; two seats.	Suggested — not playable
CAT23 — Colour Spectrum Duel	Entrants claim non-overlapping byte-mapped colour bands. A claimed-band result wins; an unclaimed result carries.	Non-overlapping bands.	Suggested — not playable
CAT24 — Prime vs Composite	Entrants take opposite PRIME/COMPOSITE sides. A proof-derived integer from 2 to 251 is classified; matching side wins.	Opposite sides only.	Suggested — not playable
CAT25 — Median Number Battle	Two distinct player picks and one proof number form three values. Pick nearest the median wins; equal distance splits.	Different numeric picks.	Suggested — not playable
CAT26 — Streak Survivor	Entrants take opposite coin sides. First side to produce a four-result proof-derived streak wins.	Opposite sides only.	Suggested — not playable
CAT27 — Territory Capture	Entrants claim non-overlapping sectors. Successive proof bytes capture sectors until one entrant controls a majority.	Non-overlapping sectors.	Suggested — not playable
CAT28 — Modulo Four Duel	Entrants select different remainders 0–3. Proof byte modulo four decides; an unclaimed result carries.	Different remainders.	Suggested — not playable
CAT29 — Poker High Duel	Five proof-derived cards per entrant. Standard hand category and deterministic kickers decide; exact ties split.	No pick; two seats.	Suggested — not playable

M.3 Suggested Arcade concepts

Code / game	Proposed outcome or reward rule	Production-specific requirement	v9.6 status
G7 — Plinko Drop	Proof-derived peg path lands in a published multiplier slot.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable
G8 — Mini Slots	Three proof-derived reels use published symbol weights, paylines and capped prizes.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable
G9 — Quick Keno	Player numbers are compared with a proof-derived draw using a published hit/pay table.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable
G10 — Bingo Rush	Proof-derived draws fill a 3×3 card; completing a line before the draw limit earns the published prize.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable
G11 — Treasure Hunt	Proof-shuffled tiles expose coins, multipliers, keys and traps; published stop/bust rules determine payout.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable
G12 — Memory Match	A proof-shuffled deck awards a published score-based prize for pairs found within a move limit.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable
G13 — Drop Ball	A proof-derived bounce path lands in a visible multiplier pocket.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable
G14 — Daily Trivia	Timed correct answers award published streak rewards and leaderboard points; question integrity needs managed content.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable

G15 — Fishing Reel	A proof-derived rarity table determines catch, collection progress and published coin reward.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable
G16 — Penalty Shootout	Five shot choices resolve against proof-derived goalkeeper directions; final score determines the published reward.	Reviewed probability/pay table, limits, server RNG/content where applicable, telemetry, abuse controls, accessibility and approval.	Suggested — not playable

M.4 Promotion gate for any roadmap game

52. Add a complete player rules screen and pre-wager disclosure; implement every winner, tie, split, carry, cancellation and error path.
53. Add deterministic tests for rule outcomes and ledger-conservation tests for stake, escrow, payout, fee, carry and Admin earnings.
54. Add Player and Admin history/detail rendering, searchable/filterable Admin data with pagination above 20 rows, and JSON/export coverage.
55. Update the Feature Directory record from Suggested only after the player route, runtime flow and audit evidence exist; use Partial where a production service is still absent.
56. Synchronize Player What's New, Admin consistency details, feature register, changelog, project documentation and audit fingerprints in the same release.

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Appendix N — v9.7 Active Bot Network, Progression and Shop Expansion

Version 9.7 expands autonomous activity, player-network growth, progression rewards and shop inventory while preserving the local play-coin demonstration boundary. Player, Admin, Feature Directory, Feature Register, changelog and audit report use the same current feature description.

N.1 Autonomous bot activity

- Normal background cadence increases from six to eight Coin Toss matches and from six to eight Catalog matches per cycle. Turbo retains its stress-test override.
- Each normal cycle also schedules two bot Arcade+ sessions across Lucky Wheel, Scratch Cards, Dice Roll, Weekly Raffle, Multiplier Ladder P2P and War Card Game.
- Bots send incoming friend requests, form bot-to-bot friendships, start lobby conversations and create visible activity events for Private Rooms, Clans and Gifting.
- Bot chat is retained in the same capped lobby stream; mute and block rules continue to apply. Bot social and Arcade logs are separately capped at 120 records.
- Bot transfers occur more frequently, and bot shopping now selects from the complete affordable non-VIP inventory across all supported cosmetic categories.
- Coin Toss, Catalog, Series Cups and tournaments remain entrant-versus-entrant; the no-real-player requirement remains valid for bot-only Cups and tournaments.

Feature family	Bot action	Balance/economy effect	Visible evidence	Production boundary
Social	Requests, friendships, chat, room/clan/gift events	No automatic player-wallet debit	Community live activity, chat and incoming requests	Authenticated identities, moderation, consent and realtime service
Lucky Wheel / Scratch / Dice	Autonomous paid sessions	Bot pays entry; published demo payout updates bot; cost/payout hits sinks/taps and promo cost	Arcade+ bot activity and Admin metrics	Reviewed mathematics and server RNG
Weekly Raffle	Bots buy 1–5 tickets	Bot balance is debited; ticket value remains in pool until 80/20 draw	Visible bot ticket count and Arcade log	Ticket ledger and scheduled draw service
Ladder / War	Bot-vs-bot P2P sessions	Both stakes debited; winner gets post-fee pot; fee recorded	Arcade log, bot stats and Admin metrics	Server-authoritative escrow and proof
Shop / Transfers	More frequent purchases and transfers	Bot pays item/transfer; shop or transfer fee recorded	Feed, bot profile inventory and Admin lists	Transactional inventory/ledger

N.2 Bot top-up and Admin controls

The bot top-up trigger is now one shared configuration value. Its default is 500 coins. This replaces earlier hard-coded 100/500 paths; the player's manual top-up minimum remains 100 and is not changed by this bot-liquidity setting.

- Admin path: Live Operations → Bot Economy & Automatic Player Growth.
- Bot top-up trigger range: 0–10,000; default 500. A bot tops up only when its current balance is below the configured value.
- Base top-ups remain varied from 400 to 1,600 and do not repeat consecutively for the same bot where the alternate value is available.
- Each bot's first top-up still receives +50% when the Admin top-up promotion is active; pausing the promotion suppresses that bonus.
- Every top-up is recorded with time, bot, base, promo, total and reason. Demo liquidity is a tap and not real payment revenue.

N.3 Automatic player-network growth

- The starting roster remains 99 bots. While the Player application is open and bots/growth are enabled, the roster automatically grows.
- Default growth: one bot every 15 seconds, up to 250 bots. Admin may set a 5–3,600 second interval, batch size 1–10 and cap 99–1,000.
- Admin can add one bot immediately. Auto-created bots receive unique names, profile metadata, balance, level, cosmetic skin and

normal game/economy fields.

- The Home live-network KPI, Player Directory, Leaderboard and Admin active-bot count all use the persisted roster length, so the displayed player count increases automatically.
- Production requires real account registration metrics to be separated from simulated liquidity bots; bot count must never be presented as human MAU/DAU.

N.4 Achievements and level rewards

The achievement catalog increases from 29 to 49. Existing unlocked achievements remain intact. New achievements unlock once and credit BONUS using the same achievement reward path.

New achievement	Trigger	BONUS reward
Friendly Five	5 friends	125
Social Circle	10 friends	250
Lobby Regular	10 player chat messages	100
Room Host	3 private rooms created	150
Gift Giver	3 gifts	150
Arcade Starter	10 Arcade+ games	150
Arcade Ace	50 Arcade+ games	500
Four-Figure Payout	Maximum payout at least 1,000	300
Volume Player	Lifetime wager at least 10,000	300
Cup Legend	10 Series Cup wins	600
Tournament Royalty	5 tournament wins	750
Wardrobe Vault	25 paid cosmetics	750
Master Level	Level 30	1000
Mythic Level	Level 40	1500
Level Summit	Level 50	2500
Community Banker	10 player transfers	300
Untouchable	15-win streak	1000
Catalog Veteran	50 Catalog matches	500
Clan Loyalist	5 clan games	300
All-Rounder	At least one Catalog, Series, Tournament, Friend and Arcade game	500

Default level rewards now exist for levels 2 through 50 and equal level $\times$ 50 BONUS. The level-up handler pays every crossed level if one XP award skips multiple levels. Levels 30, 40 and 50 also grant one rare, epic and legendary cosmetic respectively; each milestone gift is persisted and cannot be awarded twice.

Level range / milestone	Default BONUS	Additional reward
2–25	Level $\times$ 50	Existing reward range retained
26–29	1,300–1,450	—
30	1,500	Rare cosmetic
31–39	1,550–1,950	—
40	2,000	Epic cosmetic
41–49	2,050–2,450	—
50	2,500	Legendary cosmetic

Because the Admin Level Reward Manager now contains 49 rows, it provides text filtering, level/reward sorting and 20-row pagination. “Save visible rewards” applies edits on the currently rendered page; Add Level creates the next configurable level.

N.5 Shop expansion

The shop grows from 96 to 134 items: 38 additions across all nine existing categories. Prices use MAIN coins, VIP discounts remain active, purchases remain permanent, and equipped-item behavior is unchanged.

Category	Added	New items
Coin skins	6	Obsidian; Solar Flare; Frostfire; Quantum Shift; Jade Fortune; Lunar Halo
Flags	4	Singapore; South Africa; New Zealand; Portugal
Personas	4	Alien; Fox; Panda; Unicorn
Frames	3	Emerald Orbit; Cosmic Halo; Crown Aura
Chat colours	4	Pink; Cyan; Crimson; Pearl
Victory FX	4	Cherry Blossom; Thunder Storm; Aurora Burst; Trophy Shower
Table themes	4	Desert Gold; Ice Palace; Royal Hall; Lava Chamber
Sound packs	3	Synthwave; Crystal Chime; Deep Bass
Premium emojis	6	Rocket; Trophy; Robot; Alien; Fox; Panda

N.6 Validation and production boundary

- Validate JavaScript syntax, player/admin boot, bot growth, social/Arcade activity, configurable threshold, top-up promotion, achievement count, level reward keys, milestone gifts, shop count and Admin pagination.
- Validate zero duplicate IDs, matching navigation panels, balanced CSS, DOCX ZIP/open integrity and live HTTP delivery.
- All bot identities and activity remain local simulation. Production must use authenticated users, server-authoritative ledgers, independently verifiable RNG, moderation, rate limits, anti-abuse controls and honest separation of bot and human analytics.

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Appendix O — v9.8 Remaining Features Phase 1: Game Expansion

Phase 1 implements all 22 previously Suggested game records. Twelve become P2P Catalog games and ten become Arcade+ games. Existing v9.7 bot activity, progression, shop, economy and Admin controls remain active.

O.1 Shared integration requirements

- CAT18–CAT29 use the same two-entrant P2P architecture as the supplied catalog: stake escrow, per-game waiting room, manual take, automatic bot join, bot-posted bets, bot-vs-bot settlement and no vs-house Catalog outcome.
- Every Catalog result records the post-fee payout, player and opponent picks, human-readable winner detail, retained merged-hash proof, per-game player history, Catalog log and Admin result/earnings visibility.
- G7–G16 expose a Player control, pre-play rule/payout disclosure, proof-derived result, MAIN stake, payout, delta, general game record, Arcade history, statistics and house accounting.
- Autonomous bot Arcade activity now selects from all 16 implemented Arcade+ modes. Bot balances remain finite/nonnegative and P2P modes debit both stakes.
- Player/Admin deep links open the exact Catalog or Arcade sub-feature. Directory status totals become 63 Implemented, 6 Partial and 8 Suggested.

O.2 Implemented Catalog games CAT18–CAT29

Code / game	Player input	Winner determination	Demo integration
CAT18 — Rock Paper Scissors Duel	ROCK / PAPER / SCISSORS	Standard RPS dominance; same choice splits.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT19 — Closest to 21	No pick; two cards each	Closest total ≤ 21 wins; equal totals or both bust split.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT20 — Triple Coin Majority	HEADS / TAILS	Majority of three proof flips wins.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT21 — Sequence Builder	HH / HT / TH / TT	First different two-symbol starter to appear wins; none carries.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT22 — Dice Sum Duel	No pick; two dice each	Higher two-dice sum wins; equal sums split.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT23 — Colour Spectrum Duel	Non-overlapping byte bands	Result inside a claimed band wins; unclaimed band carries.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT24 — Prime vs Composite	PRIME / COMPOSITE	Proof integer 2–251 is classified; matching side wins.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT25 — Median Number Battle	Distinct values 0–255	Pick at the median wins; if proof is median, nearest pick wins; equal distance splits.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT26 — Streak Survivor	HEADS / TAILS	First side to four consecutive proof flips wins; no streak carries.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT27 — Territory Capture	Non-overlapping sectors	Nine proof bytes capture sectors; higher count wins; equal nonzero splits; zero-zero carries.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT28 — Modulo Four Duel	Different remainders 0–3	Proof byte modulo four matching a claimed remainder wins; unclaimed result carries.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result
CAT29 — Poker High Duel	No pick; five cards each	Standard category and deterministic kickers decide; exact ties split.	Waiting + manual take + auto bot + bot-vs-bot + proof + history + Admin result

O.3 Implemented Arcade+ games G7–G16

Code / game	Control	Published reward rule	Recorded output
G7 — Plinko Drop	No choice	Nine slots: 0×, 0.5×, 1×, 2×, 5×, 2×, 1×, 0.5×, 0×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting
G8 — Mini Slots	No choice	Pair 2×; triple 5×; triple 7 pays 10×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting

G9 — Quick Keno	Five distinct numbers 1–20	Five-number draw: 2 hits 0.5×, 3 hits 1.5×, 4 hits 5×, 5 hits 20×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting
G10 — Bingo Rush	No choice	Proof-shuffled 3×3 card and six draws; any row/column/diagonal pays 3×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting
G11 — Treasure Hunt	Tile 1–9	Proof-shuffled traps/coins/chests/key; published 0×–5× table, key 4×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting
G12 — Memory Match	No choice	Eight proof-driven moves: 2 pairs 1×, 3 pairs 2×, 4 pairs 4×, 5+ pairs 8×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting
G13 — Drop Ball	Column 1–7	Seven bounces reach pockets 0×, 0.5×, 1×, 3×, 1×, 0.5×, 0×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting
G14 — Daily Trivia	One answer	One local attempt per day; correct answer pays 3×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting
G15 — Fishing Reel	Basic / Shiny / Legend bait	Common 0.5×, Uncommon 1.2×, Rare 3×, Legendary 10×; collection retained.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting
G16 — Penalty Shootout	LEFT / CENTER / RIGHT	Five kicks: 3 goals 1.5×, 4 goals 3×, 5 goals 8×.	Proof, stake, payout, delta, game history, Arcade history, statistics and accounting

O.4 State, history and accounting

- Extended Arcade state is stored under featureGames.extended with capped plays, per-game latest result, Daily Trivia date and Fishing collection counts.
- Direct Arcade stakes use the existing feature-spend/shop-revenue and sink path. Coin payouts update taps and promotional payout cost. Invalid Keno input is rejected before debit.
- Catalog fees continue through catalogFees and jackpot funding. Carries remain isolated per game ID and are paid only when that game next settles.
- All new player games write the shared capped game list. Catalog games additionally write catalogLog; G7–G16 additionally write Arcade history.
- Feature Directory links use tab plus feature query parameters; the Player deep-link handler selects the requested Catalog or Arcade game.

O.5 Validation and remaining phases

- Validate 29 Catalog definitions, 12 new resolver branches, one new P2P settlement conservation path, 10 Extended Arcade play paths and all 16 bot Arcade modes.
- Validate Directory total/unique codes and status counts, deep links, syntax, boot, IDs, navigation, CSS, DOCX package and live HTTP delivery.
- Phase 2 will address the remaining Platform, Security and Responsible Gaming Suggested/Partial records. These production-like controls remain demo simulations until backed by authenticated services.

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Appendix P — v9.9 Remaining Features Phase 2: Trust Center

Phase 2 implements working local-demo flows for all 14 records that remained Suggested or Partial after the v9.8 game expansion. Three features retain Partial status only because a static browser cannot supply their required production backend: Public API, remote Push Notifications and account-enforced Two-Factor Authentication.

P.1 Platform features

Code / feature	Working demo	Production boundary	Directory status
T1 — Installable PWA	manifest.webmanifest, theme/mobile metadata, 192/512 PNG icons, service-worker registration, offline core cache and install-readiness action	A secure origin and browser install policy are still required.	Implemented
T2 — Multi-language	Persistent English, Hindi, Bengali, Tamil and Telugu primary navigation and Home messaging	Full production localization still requires translated content review and locale testing.	Implemented
T3 — Public API	In-browser status, leaderboard, results and fairness explorer; demo-key rotation; request log; static OpenAPI 3.0 file	Partial: authenticated server routes, rate limits and key custody are required.	Partial
T4 — Realtime Analytics	Five-second capped samples for visible players, games, gross revenue, queue, social and Arcade; Admin KPIs, chart and JSON export	Production needs an event pipeline, durable warehouse and bot/human separation.	Implemented
T5 — Push Notifications	Browser permission, match/friend/jackpot/quest preferences, Notification API/local fallback and alert history	Partial: closed-app delivery needs a remote push/subscription service.	Partial

P.2 Security and statements

Code / feature	Working demo	Production boundary	Status
SEC1 — Two-Factor Authentication	Generate Base32 secret; Web Crypto HMAC-SHA1 six-digit TOTP; verify current $\pm$ one 30-second window; verified disable	Partial: protected account secret custody and server enforcement required.	Partial
SEC2 — Anti-Cheat Detection	Player/Admin rules scan impossible balances, extreme win rate, duplicate proof, malformed payout and unsafe Turbo; high findings enter review flags	Production requires server events, behavior models, case management and appeal process.	Implemented
SEC3 — Public Fairness Page	Existing standalone proof recomputation remains available without account access	Production commitment service and immutable proofs required.	Implemented
SEC4 — Monthly Activity Statement	Month filter, preview, wagers/payouts/fees/P&L/top-ups/transfers, JSON/CSV downloads and simulated email log	Production requires authoritative ledger, PDF/email delivery and data-retention controls.	Implemented

P.3 Responsible Gaming

Code / feature	Enforcement in demo	Production boundary	Status
RG1 — Deposit Limits	Persist daily/weekly/monthly limits and deposits; enforce before manual top-up; decreases/new restrictions immediate; increases/removal delayed 60 seconds in demo	Production delay should be policy-defined (for example 24 hours) and server-enforced.	Implemented
RG2 — Session Time Limits	Persist maximum session and cool-off minutes; check before every new game; reaching limit starts cool-off and blocks play	Server/session authority required across devices.	Implemented
RG3 — Durable Self-Exclusion	Player selects 1-minute demo, 24 hours, 7 days or permanent; shared-state game guard; Auto Bet disabled; no Player/Admin early undo	Production needs identity-level exclusion, regulator/operator integration and protected records.	Implemented
RG4 — Reality Check Graph	Configurable 1/5/10/15-minute reminders, capped session P/L points, graph and explicit up/down callout	Production should use server time and accessible modal acknowledgement.	Implemented
RG-D — Combined demo controls	Existing loss limit, bank and Auto Bet stop now coexist with RG1–RG4 in one Player screen and Admin posture view	Server authority remains required.	Implemented

P.4 Promotion redemption completion

- Admin campaigns now receive a stable ID and retain type, amount, start, end, enabled state and creation time.
- Credit and cash-drop campaigns can be claimed once from Player Safety & Services. Reward, tap and promotion cost are recorded.
- Deposit-percent campaigns activate for the next manual top-up. The bonus is credited to BONUS, the campaign is marked claimed and cannot be reused.
- Admin campaign cards and Trust Center show Player claim counts. Pause, resume, schedule, expiry and deletion remain available.
- The original first-top-up promotion remains independent and can still be paused by Admin.

P.5 State, screens and navigation

- Player adds a nineteenth navigation tab: Safety & Services, with Platform, API, Security, Responsible Gaming, Statements and Offers sub-screens.
- Admin adds an eleventh navigation tab: Trust Center, with analytics, anti-cheat, RG posture, platform/account service status and claim/statement metrics.
- Feature Directory deep links open the exact Player service or Admin Trust Center. All 77 codes remain unique.
- New shared state namespaces are services, rg and analytics. Existing v9.8 saves migrate through deep defaults.
- PWA support files are manifest.webmanifest, sw.js, icons/icon-192.png, icons/icon-512.png and api/openapi.json.

P.6 Phase 2 status and validation gate

Status	Count	Meaning
Implemented	74	All complete local-demo flows, including production-boundary labels
Partial	3	T3 Public API, T5 Push Notifications and SEC1 2FA require remote/account services
Suggested	0	No unimplemented Feature Directory roadmap records remain
Total	77	Unique codes

- Validate Player/Admin syntax and boot, 19/11 navigation-panel matching, zero duplicate IDs, CSS balance and version labels.

- Exercise API endpoints, notification log, TOTP verify, anti-cheat, deposit/session enforcement, delayed limit increase, self-exclusion, analytics sample, statement and both promotion claim types.
- Validate manifest/OpenAPI JSON, service-worker syntax, icon dimensions, DOCX package, fingerprints and live HTTP delivery.

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Appendix Q — v10.0 Player and Admin Navigation Accessibility

Version 10.0 restructures access without removing or merging screens. Player Directory and Leaderboard remain separate. All existing deep links, Feature Directory destinations and sub-feature links continue to work.

Q.1 Player navigation

Control	Behavior
Desktop sidebar search	Filters all 19 destinations by visible label, screen name, description and keywords. Group labels collapse during search; a no-results message is displayed.
Grouped quick jump	Direct select access to Coin Toss, Series/Tournaments, Catalog, Arcade+, Wallet, History, Statistics and Safety & Services.
Command palette	Searchable grouped list of all 19 screens with icon, name and purpose. Open with header button, Ctrl/Command+K or ` / `; close with Escape, close button or backdrop.
Mobile dock	Fixed Home, Play, Catalog and Wallet actions plus More. More opens the full command palette so every non-dock destination remains one search away.
Synchronized state	Navigating by sidebar, quick jump, command result, mobile dock, Feature Directory or deep link updates active panel, quick-jump value and dock highlight.

Q.2 Admin navigation

Control	Behavior
Desktop sidebar search	Filters all 11 Admin destinations using screen label, description and operational keywords.
Grouped quick jump	Fast access to Overview, Live Operations, Feature Directory, Trust Center, Rates, Economy, Promotions and Audit.
Command palette	Grouped Command, Commercial, Engagement and Governance destinations with Ctrl/Command+K, ` / `, Escape and backdrop controls.
Mobile dock	Fixed Overview, Live Operations, Feature Directory and Trust Center actions plus More for all Admin destinations.
Synchronized state	Admin sidebar, jump select and dock highlight update together after any navigation path.

Q.3 Accessibility and responsive behavior

- Command overlays use dialog and aria-modal semantics, accessible labels and explicit close controls.
- All search and quick-jump inputs have aria labels. Search results use full-width keyboard-focusable buttons with visible focus/hover states.
- Keyboard shortcuts do not intercept ` / ` while focus is inside an input, textarea or select.
- Desktop navigation tools hide at tablet/mobile breakpoints to preserve the existing horizontal screen tabs; the fixed mobile dock supplies primary actions.
- Body bottom padding prevents the mobile dock from covering Player/Admin content.
- The command palette is responsive up to 96 viewport-width, limits its height and scrolls results independently.

Q.4 Destination inventory

Application Count Destinations

Player	19	Home; Live Lobby; Coin Toss; Series & Tournaments; P2P Catalog; Arcade+; Leaderboard; Players; Social Hub; Progress+; Economy+; Shop; VIP & Season; History; What’s New; Wallet; Statistics; Fairness; Safety & Services
Admin	11	Overview; Live Operations; Feature Hub; Feature Directory; Rates & Jackpot; Economy; Promotions; VIP & Levels; Tournaments; Audit & Data; Trust Center

Q.5 Validation gate

- Validate Player/Admin boot at version 10.0 and exact navigation metadata counts of 19 and 11.
- Exercise command search/open/close, sidebar filtering, quick-jump selection, direct tab navigation, active-state synchronization and mobile More behavior in both apps.
- Validate matching tab/panel sets, zero duplicate IDs, literal ID references, CSS balance, JavaScript syntax, DOCX integrity and live HTTP delivery.
- Confirm all v9.9 Feature Directory statuses and deep links remain unchanged.

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Appendix R — v10.1 Catalog and Arcade+ Game Navigation

Version 10.1 applies the same easy-access model introduced in v10.0 to the two longest in-screen game lists. No game rule, wallet path, waiting room, proof, history or bot behavior is removed.

R.1 P2P Catalog navigation

Control	Behavior
Search	Name, CAT code, rule description, edge and category; updates the visible tab list and result count.
Category	All; Favorites; Side Picks; Numbers & Dice; Patterns & Territory; Cards.
Quick jump	Grouped select containing all 29 games; selecting a game opens its existing panel.
Favorites	Current-game star toggle persists the game ID in shared settings; Favorites category filters to saved games.
Result count	Shows visible/total games plus active category and Favorites-only state.
Command access	Player command palette includes all 29 exact games. Admin command palette opens exact Player game deep links.

R.2 Catalog categories

Category	Count	Games
Side Picks	12	Over/Under, Speed, Tug, Even/Odd, Blind, Chain, Ladder, Mirrored, RPS, Triple Coin, Prime/Composite, Streak Survivor
Numbers & Dice	8	Closest, Lucky Battle, Sum Prediction, Higher Byte, Closest 21, Dice Sum, Median Battle, Modulo Four
Patterns & Territory	8	Pattern Race, Parlay, Prediction Streak, Range War, Bullseye, Sequence Builder, Colour Spectrum, Territory Capture
Cards	1	Poker High Duel

R.3 Arcade+ navigation

Control	Behavior
Search	Game title, G-code, rule description and category; updates the visible sub-tab list and result count.
Category	All; Favorites; Classic & Chance; P2P Duels; Puzzle & Picks; Daily & Collection.
Quick jump	Grouped select containing all 16 modes; selection opens the existing Arcade flow.
Favorites	Current-game star toggle persists the Arcade key in shared settings.
Result count	Shows visible/total games and active filter state.
Command access	Player/Admin global palettes provide exact deep links for G1–G16.

R.4 Arcade+ categories

Category	Count	Games
Classic & Chance	6	Lucky Wheel, Scratch Cards, Dice Roll, Plinko, Mini Slots, Drop Ball
P2P Duels	2	Multiplier Ladder, War Card
Puzzle & Picks	5	Quick Keno, Bingo Rush, Treasure Hunt, Memory Match, Penalty Shootout
Daily & Collection	3	Weekly Raffle, Daily Trivia, Fishing Reel

R.5 Responsive and accessibility behavior

- Toolbars use labeled native search/select controls and responsive four/two/one-column layouts.
- At narrow widths, search, category, quick jump and Favorite stack vertically with a full-width action button.
- Empty filters display explicit no-match messages rather than a blank tab strip.
- Favorite stars appear in visible Catalog/Arcade tabs and are not used as the only indication; the Favorite button text also changes.
- Global command results include game icon, name, code/rule description and exact destination.
- Existing Feature Directory and query-parameter deep links remain compatible.

R.6 Validation gate

- Validate Catalog group inventory $12/8/8/1 = 29$ and Arcade group inventory $6/2/5/3 = 16$.
- Exercise search, category, Favorites, grouped quick jump, favorite persistence, visible-result count and active-game synchronization.
- Exercise Player/Admin command searches for exact Catalog and Arcade games and confirm deep-link targets.

- Validate v10.1 boot, tab/panel mapping, IDs, CSS, syntax, DOCX integrity and live delivery.

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Appendix S — v10.2 Cons Review and New-Feature Roadmap

Version 10.2 performs a product and production-readiness review after the v10.1 navigation release. It does not present Suggested ideas as implemented. The authoritative detailed register is TossMatch_Cons_and_Roadmap.md.

S.1 Principal cons and disadvantages

Severity	Con	Consequence	Correction / roadmap
Critical	Client-authoritative wallet, settlement, promotions and RG	Balances, outcomes, limits and claims can be changed through browser state	Server-authoritative transactional ledger and immutable events
Critical	No protected Admin/account service	Admin state, TOTP and API keys are not identity-bound	Authentication, RBAC, MFA, approval workflows and session control
High	Client-generated fairness secret	Commit/reveal is educational rather than independently trustworthy	Independent commitment/reveal service and external verification
High	Unsafe HTML interpolation	Networked user text could inject markup/script	Central encoding/sanitization, textContent and CSP
High	localStorage persistence	Mutable, clearable, cross-tab conflicts and no cross-device enforcement	Database, migrations, optimistic locking and recovery
High	Local RG enforcement	Limits/exclusion can be bypassed across browser/device state	Identity-level server RG service
High	Uncertified game mathematics	Not suitable for regulated or real-value use	Independent math, fairness and security certification
High	No identity/age/jurisdiction eligibility	No age gate, KYC or geofencing	TRUST1
Medium	Monolithic Player/Admin HTML	Higher regression and maintenance risk	Modular TypeScript and schemas
Medium	Incomplete browser/accessibility automation	DOM stubs cannot validate all real-browser behavior	Playwright, axe-core and UX1
Medium	Partial translation	Primary navigation is translated, not all rules/content	Professional localization workflow
Medium	Bot/human metrics blended	Simulated growth may be mistaken for human activity	Separate analytics dimensions and public labels
Medium	Histories capped and mutable	Weak authoritative replay/dispute evidence	LIVE2 and TRUST3
Medium	No privacy/dispute/device/incident centers	Trust workflows are not actionable	TRUST2, TRUST3, TRUST4 and OPS5
Low	Fixed Home layout and repeated setup	Less personalization and repeated input	UX2 and UX3
Low	Discovery and event schedule are fragmented	45 games/events can be difficult to discover	UX4 and LIVE1

S.2 Suggested new features

Code	Priority	Suggested feature	Proposed behavior
UX1	P1	Accessibility Center	High contrast, reduced motion, text scaling, colour-blind and screen-reader preferences
UX2	P3	Customizable Dashboard	Selectable/reordered KPI cards and persisted layouts
UX3	P2	Saved Bet & Game Presets	Named stake/game/Auto Bet presets with confirmations and RG checks
UX4	P3	Smart Game Discovery	Explainable recommendations based on favorites, recency, variety and limits
LIVE1	P2	Event Calendar & Scheduled Play	Timezone-aware tournaments, Cups, promotions and reminders
LIVE2	P2	Match Replay & Shareable Proof	Step replay, redacted proof package and share link
LIVE3	P3	Clan Seasons & Cooperative Quests	Divisions, shared goals, contribution ledger, rewards and archives
TRUST1	P1	Identity, Age & Jurisdiction Checks	Age gate, KYC status, geofencing and feature eligibility
TRUST2	P1	Privacy & Data Rights Center	Consent, export, correction, deletion and retention workflows

TRUST3 P1	Dispute & Support Case Center	Case creation from game/transaction, proof, SLA and resolution
TRUST4 P1	Device & Session Management	Active sessions, revoke, unusual access and step-up verification
OPS5 P2	Status & Incident Center	Service health, incident timeline, notices and postmortems

S.3 Recommended architecture work

- Server-authoritative ledger, escrow, RNG commitments, promotion claims and responsible-gaming enforcement.
- Authenticated account/Admin services with RBAC, MFA and device/session controls.
- Modular TypeScript packages, schema migrations and transactional database persistence.
- Deterministic game-rule and ledger-conservation tests plus long-running bot/Turbo tests.
- Playwright mobile/desktop/PWA coverage, axe-core accessibility and professional localization QA.
- Central output encoding, CSP, rate limiting, abuse controls and independent audits.
- Observability pipeline with bot/human dimensions, case management and incident response.

S.4 Prioritized delivery sequence

Stage	Features	Purpose
Foundation	UX1, TRUST1, TRUST2, TRUST3, TRUST4	Accessibility and trust/compliance baseline plus server-account architecture
LiveOps & transparency	UX3, LIVE1, LIVE2, OPS5	Presets, event schedule, replay/proof sharing and public status
Engagement	UX2, UX4, LIVE3	Personal dashboard, explainable discovery and clan seasons

S.5 Directory and status boundary

- Feature Directory totals: 74 Implemented, 3 Partial/demo-only, 12 Suggested and 89 unique records.
- Suggested entries have no Player/Admin quick link and render as Roadmap item.
- No Suggested feature may move to Implemented until a screen, runtime flow, persistence, tests and synchronized documentation exist.
- The production-service Partial boundary remains T3 Public API, T5 remote Push Notifications and SEC1 account-enforced TOTP.

□

Appendix T — v10.3 Demo UX Improvements

The scope decision is explicit: fix dashboard layout and repeated game setup in the demo, implement the four UX proposals, and defer every other production risk/feature until production infrastructure exists. Existing warnings are retained rather than marked resolved.

T.1 UX1 Accessibility Center

- Player path: Safety & Services → UX Lab.
- Persistent controls: high contrast, reduced motion, screen-reader hints, text scale 90/100/110/120/130%, and Standard/Deuteranopia/Protanopia/Tritanopia colour presets.
- Apply is immediate. Reset restores standard contrast/motion, 100% text and screen-reader hints.
- An aria-live region announces active accessibility scale when hints are enabled.
- This is a user preference demo; professional localization, screen-reader certification, focus-trap and WCAG browser testing remain production work.

T.2 UX2 Customizable Dashboard

KPI card	Persisted ID
Playable wallet	wallet
Career net	net
Jackpot pool	jackpot
Level / XP	level
Live player network	network
Lifetime wagered	wagered
Maximum payout	payout
Arcade+ plays	arcade

Players can show/hide and reorder KPI cards. The four major Home sections—Live Opportunities, VIP Journey, Activity Pulse and Explore TossMatch—can also be shown/hidden and reordered. At least one item in each list must remain visible. Reset restores five default KPIs and all four sections.

T.3 UX3 Saved Bet & Game Presets

Type	Stored fields	Apply behavior
Coin Toss	Name, stake, HEADS/TAILS, current Auto Bet stop	Opens Coin Toss, selects side and fills stake
Catalog	Name, stake, exact Catalog game, current Auto Bet stop	Opens exact Catalog game and fills stake

Arcade+	Name, stake, exact Arcade game, current Auto Bet stop	Opens exact Arcade game and fills available stake control
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- Up to 20 presets are retained in shared settings; newest appears first.
- Preset application never clicks Play/Post and never bypasses escrow, balance, loss, deposit, session or self-exclusion guards.
- Player receives an explicit review-and-confirm message after applying a preset.
- Presets can be deleted individually.

T.4 UX4 Smart Game Discovery

- Recommendations are explainable and never auto-wager.
- Sources: saved Catalog favorites, saved Arcade favorites, an unplayed Catalog variety pick, current Catalog game, Daily Trivia availability/completion and Fishing collection opportunity.
- Each recommendation displays its reason and opens the exact Catalog/Arcade feature.
- Production recommendations should use account-backed data, fairness controls and responsible-gaming policy review.

T.5 Deferred production scope

Deferred	Production scope
LIVE1–LIVE3	Event calendar, replay/shareable proof, clan seasons/cooperative quests
TRUST1–TRUST4	Identity/age/jurisdiction, privacy/data rights, disputes/support, device/session management
OPS5	Status and incident center
Architecture/security cons	Server ledger/RG, Admin/account security, independent fairness, safe rendering, durable persistence, certification and full browser accessibility

T.6 Status and validation

- Feature Directory: 78 Implemented, 3 Partial/demo-only, 8 Suggested, 89 unique records.
- Validate accessibility apply/reset, dashboard hide/show/reorder/reset, preset save/apply/delete and recommendation reasons/exact links.
- Validate UX1–UX4 Player links and remaining eight roadmap records without links.
- Validate syntax, boot, IDs, navigation, CSS, DOCX, fingerprints and live delivery.

□

Appendix U — v10.4 Coin-Use Expansion

Version 10.4 implements the selected balanced expansion: four P2P Catalog games, four Arcade+ games and four non-game coin utilities. All balances are play coins. Existing production warnings and Suggested features remain unchanged.

U.1 CAT30–CAT33 P2P Catalog games

Code / game	Input	Winner / carry rule	Integration
CAT30 — Three Dice Poker	No pick; three dice each	Triple beats pair, then higher total/high die; exact tie splits	Escrow, waiting, manual take, auto bot, bot-vs-bot, fee, proof, history and Admin results
CAT31 — Last Digit Duel	Different digits 0–9	Proof number last digit matching a claimed digit wins; unclaimed digit carries	Escrow, waiting, manual take, auto bot, bot-vs-bot, fee, proof, history and Admin results
CAT32 — Binary Code Duel	Different H/T three-bit codes	Lowest Hamming distance to proof code wins; equal distance splits	Escrow, waiting, manual take, auto bot, bot-vs-bot, fee, proof, history and Admin results
CAT33 — Coin Balance Battle	Different HEADS-count predictions 2–8	Closest to HEADS count in ten proof flips wins; equal distance splits	Escrow, waiting, manual take, auto bot, bot-vs-bot, fee, proof, history and Admin results

U.2 G17–G20 Arcade+ games

Code / game	Control	Published payout	Recorded output
G17 — Coin Pusher	No choice	Five proof drops produce a published 0×–6× tray outcome	Stake, payout, delta, proof, game history, Arcade history, statistics and accounting
G18 — Tower Builder	Target floor 3, 5 or 7	Every floor risk check must pass; target pays 1.6×, 3× or 7×	Stake, payout, delta, proof, game history, Arcade history, statistics and accounting
G19 — Match-3 Rush	No choice	Proof 5×5 board; 2/4/6+ matches pay 1.5×/4×/10×	Stake, payout, delta, proof, game history, Arcade history, statistics and accounting
G20 — Mystery Vault	Key 1–5	Correct proof-selected key pays 4.5×	Stake, payout, delta, proof, game history, Arcade history, statistics and accounting

U.3 E6–E9 non-wager coin utilities

Code / utility	Coin use	Accounting	Safety boundary
			MAIN only; retained history; all

E6 — Cosmetic Crafting	150/400/800 MAIN uncommon/rare/epic recipes	House/coin sink; grants rarity-floor cosmetic; crafting history	existing RG/wallet rules remain active
E7 — Event Ticket Packs	1 for 100, 3 for 250, 7 for 500 MAIN	House/coin sink; persistent non-payout demo utility tokens	MAIN only; retained history; all existing RG/wallet rules remain active
E8 — Clan Treasury	Minimum 50 MAIN contribution	Non-withdrawable social resource and sink; no house revenue, withdrawal or wagering advantage	MAIN only; retained history; all existing RG/wallet rules remain active
E9 — Private Room Upgrades	Basic free; Neon 200; Royal 500; Cosmic 1,000	Visual-only persistent room tier; no odds, payout, queue or matchmaking advantage	MAIN only; retained history; all existing RG/wallet rules remain active

U.4 Bot, Admin and navigation integration

- Autonomous bot Arcade mode selection derives from the extended registry and now includes all 20 modes.
- Admin Feature Hub reports utility crafts, tickets, treasury and room tier.
- Feature Directory links open CAT30–CAT33, G17–G20 and E6–E9 exact Player features.
- Player command palette contains 19 screens + 33 Catalog + 20 Arcade = 72 entries.
- Admin command palette contains 11 Admin screens + 33 Catalog + 20 Arcade = 64 entries.
- Catalog category totals become 12 Side Picks, 11 Numbers & Dice, 9 Patterns & Territory and 1 Cards.
- Arcade category totals become 8 Classic & Chance, 2 P2P Duels, 7 Puzzle & Picks and 3 Daily & Collection.

U.5 Status and validation

- Feature Directory: 90 Implemented, 3 Partial/demo-only, 8 Suggested, 101 unique records.
- Validate four new Catalog resolvers and a valid player/bot/split/carry result detail.
- Complete all four new Arcade player flows with proof, history and finite balances.
- Exercise all four utility spends and verify wallet deductions, state and history.
- Validate syntax, IDs, navigation, CSS, DOCX, fingerprints and live delivery.

□

TossMatch • Play-coins demonstration •

TOSSMATCH • PROJECT DOCUMENTATION • v9.6 DEMO

Appendix V — Consolidated Authoritative v10.4 Specification

Status: AUTHORITATIVE CURRENT-STATE CONSOLIDATION — 25 August 2026. This appendix makes the DOCX self-contained for the complete v10.4 Player/Admin rules and feature set. When an earlier release appendix or summary conflicts with current behavior, this Appendix V controls. The synchronized Markdown source is TossMatch_Complete_Feature_and_Rules_Specification.md; index.html and admin.html remain the executable source of truth.

Authoritative documentation revision: 25 August 2026

Runtime: Local browser play-coin demonstration

Release counts: 33 P2P Catalog + 20 Arcade+ = 53 game destinations; 72 Player commands; 101 unique Feature Directory records (90 Implemented / 3 Partial / 8 Suggested).

Shared state key: tossmatch_v8

Scope: Player, Admin, 33 Catalog games, 20 Arcade+ modes, bots, wallet/economy, progression, Platform/Security/RG, UX, feature directory, cons and production boundaries.

1. Deliverables and version

Deliverable	Purpose
index.html	Player v10.4 application
admin.html	Admin v10.4 control panel
manifest.webmanifest	Installable PWA metadata
sw.js	Offline core-cache service worker
icons/icon-192.png / icon-512.png	PWA icons
api/openapi.json	Demo API specification
TossMatch_Project_Documentation.docx	Complete project documentation
TossMatch_Feature_Register.md	Feature-by-feature status
TossMatch_Cons_and_Roadmap.md	Cons, production deferrals and roadmap
TossMatch_Audit_Report.md	Validated build findings/fingerprints
CHANGELOG.md	Release chronology

2. Non-negotiable defaults and rules

Setting	Current value / behavior
Default player MAIN balance	1,000
Initial BONUS / REFERRAL	250 / 50
Bot starting roster	99; grows automatically when enabled
Bot top-up trigger	500 by default; Admin range 0–10,000
Bot top-up values	Varied 400–1,600; first eligible top-up receives +50% if promotion enabled
Coin/Catalog fee	5% default
Cup rake	5% default
Tournament rake	10% default
Jackpot funding	10% of fee; floor 1; armed at 50; pays 50% of pool
Non-MAIN stake cap	20% of stake
Transfer	2% fee; minimum 10; default daily cap 500
Auto Bet stop	–200 default; presets –100/–200/–500; custom –50 to –10,000
Admin data paging	20 rows when a dataset exceeds 20
Play mode	Play coins only; no deposits/withdrawals/payment custody

3. Wallet and settlement

Segment	Use
MAIN	Winnings, manual top-ups, transfers and primary spend
BONUS	Promotions, level/achievement/quest rewards; capped non-MAIN stake contribution
REFERRAL	Referral fee share; capped non-MAIN stake contribution
RAKEBACK	Claimed VIP rakeback; capped non-MAIN stake contribution
BANK	Player-controlled parked MAIN balance; unavailable for betting until unparked

Before each wager, maintenance, durable self-exclusion, cool-off, session limit, rate limit, loss limit, minimum/maximum stake, total balance and MAIN/non-MAIN requirements are checked. P2P stakes are debited into local escrow before outcome resolution. Split/carry behavior is game-specific and every retained result stores stake, fee, payout, delta, opponent, picks/details and proof where applicable.

4. Coin Toss, Series and tournaments

Mode	Rules / winner determination
Coin Toss	Two entrants choose opposite HEADS/TAILS. Combined SHA-256 proof first byte parity decides. Both stakes enter escrow; winner receives post-fee pot. Byte 00 may trigger an armed jackpot.
Series Bo3	First entrant to 2 wins receives post-rake pot.
Series Bo5	First to 3 wins.
Series Bo7	First to 4 wins.
Series Advantage	Win by two; capped at nine games.
Tournament single	4/8/16-player single-elimination; one proof flip per match; bots fill all seats; no real player required.
Tournament Bo3	Each bracket match is first to two; bots fill and run unattended.
Tournament prizes	Configured rake removed; remaining pool uses 75% champion / 25% runner-up allocation. VIP may discount player entry.

4.1 Coin Toss controls, matching and fairness proof

Detail	Current behavior
Stake	Default 100, minimum 10, quick values 50/100/250/500/1,000; effective maximum is $\min(1,000, 100 + \text{level} \times 100)$.
Pick and posting	Choose HEADS or TAILS, debit the segmented wallet into escrow and post to the waiting room. Player can cancel an unmatched own post for full refund.
Matching	A player can manually take an opposite open bet; automatic bot matching and bot-posted liquidity also operate. Friend and Private Room Coin Toss reuse the settlement path and add context history.
Auto Bet	Persistent on/off option. Stop threshold defaults to -200 , has $-100/-200/-500$ presets and accepts -50 through $-10,000$ in steps of 50. It stops once session net is at/below the threshold.
Turbo stress modes	$1\times$, $20\times$, $100\times$, $500\times$ and $1,000\times$. Default saved state is $1\times$. Turbo affects simulated bot throughput, not payout odds; $1,000\times$ is flagged by the local anti-cheat scan as stress-mode risk.
Proof formula	$\text{makerHash}=\text{SHA256}(\text{makerSeed}:\text{gameId}:\text{stake}:\text{side})$; same for taker; $\text{commit}=\text{SHA256}(\text{serverSeed})$; add the three hashes as big integers; $\text{finalHash}=\text{SHA256}(\text{combinedHex}:\text{gameId})$; even first byte = HEADS, odd = TAILS. Byte 00 may trigger an armed jackpot.
Verification	The Player verifier accepts revealed server seed, maker hash, taker hash and game ID; "Load last game" fills the most recent proof. This is a local demonstration, not an independently hosted commitment service.

5. P2P Catalog — 33 playable games

#	Code	Game	Category	Input	Winner / tie / carry
1	Base · overunder	Over / Under	Side Picks	HIGH / LOW	Proof byte 128–255 = HIGH, 0–127 = LOW; matching side wins.
2	Base · speed	Speed Round	Side Picks	HEADS / TAILS	Five flips; majority HEADS/TAILS wins.
3	Base · tug	Tug of War	Side Picks	LEFT / RIGHT	Proof flips pull LEFT/RIGHT; first side to three steps wins.
4	Base · evenodd	Even / Odd Sum	Side Picks	EVEN / ODD	Two proof bytes are summed; parity matching EVEN/ODD wins.
5	Base · closest	Closest Number	Numbers & Dice	Number 0–255	Closest distinct 0–255 pick to proof byte wins; equal distance splits.
6	Base · luckybattle	Lucky Number Battle	Numbers & Dice	Number 0–255	Exact proof-byte match wins; otherwise closest pick; equal distance carries.
7	Base · sumpredict	Sum Prediction	Numbers & Dice	Number 0–510	Closest distinct prediction to the sum of two proof bytes wins; equal distance splits.
8	Base · higherbyte	Higher Byte	Numbers & Dice	No pick	Independent byte per entrant; higher byte wins; equal bytes split.
9	Base · patternrace	Pattern Race	Patterns & Territory	pattern3	First selected three-flip pattern to appear wins; none in proof window carries.
10	Base · parlayduel	Parlay Duel	Patterns & Territory	pattern3	Three flips; prediction with most positional matches wins; equal score carries.
11	Base · prediction	Prediction Streak	Patterns & Territory	pattern5	Five flips; most correct positional predictions wins; equal score splits.
12	Base · blind	Blind Pick	Side Picks	HEADS / TAILS	Hidden HEADS/TAILS picks. Opposite picks resolve normally; same correct picks split and same wrong picks carry.
13	Base · rangewar	Range War	Patterns & Territory	zones	Result inside a claimed non-overlapping quarter wins; outside both claims carries.
14	Base · bullseye	Bullseye	Patterns & Territory	zones	Result inside a claimed non-overlapping target wedge wins; miss carries.
15	Base · chain	Chain Reaction	Side Picks	HEADS / TAILS	Entrants alternate calls for up to ten flips; first incorrect caller loses; both surviving split.
16	Base · ladder	Elimination Ladder	Side Picks	HEADS / TAILS	HEADS/TAILS sides race; first side to three proof results wins.
17	Base · mirrored	Mirrored Coins	Side Picks	HEADS / TAILS	Two coins; matching outcomes carry; different outcomes assign one coin to each side and matching player pick wins.
18	CAT18 · rps	Rock Paper Scissors Duel	Side Picks	ROCK / PAPER / SCISSORS	Standard ROCK/PAPER/SCISSORS dominance; same choices split.
19	CAT19 · closest21	Closest to 21	Numbers & Dice	No pick	Two cards per entrant (1–10). Closest total at/below 21 wins; equal totals or both bust split.
20	CAT20 · triplecoin	Triple Coin Majority	Side Picks	HEADS / TAILS	Majority of three proof flips wins.
21	CAT21 · sequencebuilder	Sequence Builder	Patterns & Territory	HH / HT / TH / TT	First selected two-symbol H/T starter to appear wins; neither appearing carries.
					Two dice per entrant; higher sum wins;

22 CAT22 · dicesumduel	Dice Sum Duel	Numbers & Dice	No pick	equal sums split.
23 CAT23 · colourspectrum	Colour Spectrum Duel	Patterns & Territory zones		Proof byte inside a claimed non-overlapping band wins; unclaimed band carries.
24 CAT24 · primecomposite	Prime vs Composite	Side Picks	PRIME / COMPOSITE	Proof integer 2–251 is classified PRIME/COMPOSITE; matching side wins.
25 CAT25 · medianbattle	Median Number Battle	Numbers & Dice	Number 0–255	Player picks and proof byte form three values; pick at median wins; if proof is median, nearest pick wins; equal distance splits.
26 CAT26 · streaksurvivor	Streak Survivor	Side Picks	HEADS / TAILS	First HEADS/TAILS side to appear four times consecutively wins; no streak carries.
27 CAT27 · territory	Territory Capture	Patterns & Territory zones		Nine proof bytes score claimed non-overlapping sectors; higher score wins; equal nonzero score splits; zero/zero carries.
28 CAT28 · modulo4	Modulo Four Duel	Numbers & Dice	0 / 1 / 2 / 3	Proof byte modulo four matching a claimed remainder wins; unclaimed remainder carries.
29 CAT29 · pokerhigh	Poker High Duel	Cards	No pick	Five unique proof-derived cards each; standard poker category and kickers decide; exact tie splits.
30 CAT30 · threedicepoker	Three Dice Poker	Numbers & Dice	No pick	Three dice each; triple beats pair, then total/high die; exact tie splits.
31 CAT31 · lastdigit	Last Digit Duel	Numbers & Dice	0 / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9	Different digits 0–9; proof number last digit matching a claim wins; unclaimed digit carries.
32 CAT32 · binaryduel	Binary Code Duel	Patterns & Territory pattern3		Different three-bit H/T codes; lowest Hamming distance to proof code wins; equal distance splits.
33 CAT33 · coinbalance	Coin Balance Battle	Numbers & Dice	Number 2–8	Different predictions 2–8; closest to HEADS count in ten flips wins; equal distance splits.

Every Catalog game uses the same per-game waiting room, manual take, automatic bot joining, bot-posted liquidity, bot-vs-bot settlement, configured pot fee, proof retention, player per-game history and Admin Catalog result/earnings path.

6. Arcade+ — 20 playable modes

Code Mode	Control / cost	Winner or payout rule
G1 Lucky Wheel	One free daily; paid spin 50 MAIN	25/50 BONUS, 100 MAIN, cosmetic/emoji or 150 jackpot-style reward
G2 Scratch Cards	Cards cost 25/100/250	Best symbol count: 3=2×, 4=5×, 5+=10×
G3 Dice Roll	Exact double or LOW 2–6 / HIGH 8–12	Exact double 30×; ranges 1.8×; seven loses
G4 Weekly Raffle	Tickets cost 10; player and bots	One ticket wins 80%; house retains 20%; weekly reset
G5 Multiplier Ladder P2P	Equal stake vs bot	Higher proof rung wins post-fee pot; equal outcome splits
G6 War Card Game	Equal stake vs bot	Higher 2–Ace card wins; ties trigger War; unresolved tie splits
G7 Plinko Drop	Default stake 50; No choice	Slots: 0×, 0.5×, 1×, 2×, 5×, 2×, 1×, 0.5×, 0×
G8 Mini Slots	Default stake 50; No choice	Pair 2× · any triple 5× · triple 7 pays 10×
G9 Quick Keno	Default stake 50; Five distinct numbers 1–20	2 hits 0.5× · 3 hits 1.5× · 4 hits 5× · 5 hits 20×
G10 Bingo Rush	Default stake 50; No choice	Complete any row, column or diagonal within six draws to pay 3×
G11 Treasure Hunt	Default stake 50; Tile 1–9	Published tile values range from 0× trap to 5× treasure; key pays 4×
G12 Memory Match	Default stake 50; No choice	2 pairs 1× · 3 pairs 2× · 4 pairs 4× · 5+ pairs 8×
G13 Drop Ball	Default stake 50; Column 1–7	Seven visible pockets pay 0×, 0.5×, 1×, 3×, 1×, 0.5×, 0×
G14 Daily Trivia	Default stake 25; One daily answer	Correct answer pays 3× and adds a daily trivia result to history.
G15 Fishing Reel	Default stake 50; Basic/Shiny/Legend bait	Common 0.5× · Uncommon 1.2× · Rare 3× · Legendary 10×
G16 Penalty Shootout	Default stake 50; LEFT/CENTER/RIGHT	3 goals 1.5× · 4 goals 3× · 5 goals 8×
G17 Coin Pusher	Default stake 50; No choice	Tray outcomes pay 0×, 0.5×, 1×, 2×, 3× or 6×
G18 Tower Builder	Default stake 50; Floor 3/5/7	Floor 3 pays 1.6× · floor 5 pays 3× · floor 7 pays 7×; any failed floor loses.
G19 Match-3 Rush	Default stake 50; No choice	2 matches 1.5× · 4 matches 4× · 6+ matches 10×
G20 Mystery Vault	Default stake 50; Key 1–5	Correct key pays 4.5×; other keys lose.

Direct Arcade spends use MAIN and the shop/sink accounting family. Coin payouts update taps and promotional cost. G1–G4 use local demo randomization and do not expose a verifier proof package. G5–G6 derive outcomes from Web Crypto fairness bytes, but their direct Arcade history currently stores result detail rather than an exposed replay package. G7–G20 derive outcomes from the local commitment/hash helper and retain the hash/proof in the general game record. Every mode writes Arcade history and player statistics. Autonomous bots select from all 20 modes. These proof boundaries must not be represented as certified production RNG.

7. Social and community

Code Feature	Implemented detail
S1 Friends & Buddies	Friend list, bot requests and connections, online status, direct challenge and friend-first bets.
S2 Private Rooms	Invite/friend access, custom game and stake rules, details and history.
S3 Player Profiles	Stats, VIP, skill, cosmetics, achievements, form graph and recent games.
S4 Spectator Mode	Animated viewer with participants, picks, pot, fee, result and proof.
S5 Gifting	MAIN coin and paid cosmetic gifts with fee, minimum, cap and history.
S6 Lobby Chat	Text, owned emojis, autonomous bot conversation, replies, mute/block and sent history.
S7 Clans & Teams	Clan identity, roster, team matches, tournaments, leaderboard and history.

8. Progression, VIP and rewards

8.1 VIP tiers and complete benefits

Tier Name	Monthly wager threshold	Rakeback %	Benefits
1 Starter	0	0	Basic access; Daily quests; Jackpot eligibility
2 Silver	1000	4	4% rakeback; Blue chat colour; 1 free premium emoji
3 Gold	3000	6	6% rakeback; Gold avatar frame; Queue priority; 2 free premium emojis

4	Platinum	8000	8	8% rakeback; Platinum VIP coin skin; 2 free premium emojis
5	Diamond	20000	12	12% rakeback; Diamond VIP coin skin; 5% tournament discount; 3 free premium emojis
6	Black Diamond	50000	15	15% rakeback; Black Diamond animated frame; Birthday-bonus eligibility; 10% tournament discount; 10% cosmetics discount; 5 free premium emojis
7	Royal	75000	17	17% rakeback; Royal Crown title; 10% tournament discount; 20% cosmetics discount; Priority-support entitlement; 8 free premium emojis
8	Legend	100000	20	20% rakeback (maximum); Legend Prism + Eternal Flame skins; 15% tournament discount; 30% cosmetics discount; All premium emojis; Gold leaderboard name; Early-access entitlement

VIP wager counters reset by UTC month or Admin action. Pending rakeback and permanently unlocked cosmetics/emojis remain. Current-tier benefits such as discounts, queue priority, support/early-access entitlements depend on the current monthly tier.

8.2 Progress+ mechanics

Code	Exact current configuration
P1 Battle Pass	Monthly pass XP milestones: 0 / 500 / 1,500 / 3,000 / 5,000. Free BONUS: 25 / 50 / 75 / 100 / 150. Premium BONUS: 75 / 125 / 200 / 300 / 500. Premium unlock costs 500 MAIN. Subscription quest multiplier applies when eligible.
P2 Login Calendar	First visit of each local day auto-claims: days 1–6 pay 50 / 75 / 100 / 125 / 150 / 175 BONUS; day 7 pays 250 BONUS plus a cosmetic.
P3 Weekly Challenges	Win 10 games = 300 BONUS; play 5 game types = 250 BONUS; reach a 3-win streak = 200 BONUS. Progress and claims use the current calendar week.
P4 Prestige	Requires level 10+. Resets level to 1 and XP to 0; increments Prestige rank, equips/unlocks Rainbow frame and adds permanent +5% XP per rank.
P5 Skill Matching	Rookie under 10 games; thereafter Bronze <45% wins, Silver 45–54%, Gold 55–64%, Elite 65%+. Optional peer-tier preference falls back to all eligible bots.

8.3 Daily quests

ID	Goal	Target	BONUS reward
settle	Settle 3 games	3	50
win	Win 1 game	1	75
cup	Play 1 Series Cup	1	50

8.4 Achievements — 49

ID	Achievement	Trigger	BONUS reward
first	First Flip	Settle your first game	30
onfire	On Fire	3-win streak	75
unstoppable	Unstoppable	5-win streak	150
jackpot	Jackpot!	Win the jackpot	300
cup	Cup Winner	Win a Series Cup	100
champ	Champion	Win a tournament	300
bigballer	Big Baller	Stake 500+ in one game	100
nightowl	Night Owl	Play 10 games	50
frequent	Frequent Flipper	Play 25 games	100
doubledigits	Double Digits	Win 10 games	100
highroller	High Roller	Reach level 5	100
comeback	Comeback Kid	Win after two straight losses	75
century	Century Club	Settle 100 games	250
veteran	Veteran	Settle 250 games	500
winner25	Quarter Century	Win 25 games	250
winner100	Winning Machine	Win 100 games	750
streak10	Volcanic	Reach a 10-win streak	500
cup5	Cup Specialist	Win 5 Series Cups	300
trny3	Bracket Boss	Win 3 tournaments	500
catalog5	Game Explorer	Play 5 different catalog games	150
catalog17	Catalog Master	Play all 33 catalog games	750
collector5	Collector	Own 5 paid cosmetics	150
collector15	Super Collector	Own 15 paid cosmetics	400
silvervip	Silver Status	Reach Silver VIP	100
goldvip	Gold Status	Reach Gold VIP	250
profit1000	Profit Maker	Reach +1,000 career net	300
level10	Double-Digit Level	Reach level 10	300
level20	Elite Level	Reach level 20	750
transfer5	Generous	Complete 5 player transfers	150
friends5	Friendly Five	Add 5 friends	125
friends10	Social Circle	Add 10 friends	250
chat10	Lobby Regular	Send 10 lobby messages	100
rooms3	Room Host	Create 3 private rooms	150
gifts3	Gift Giver	Send 3 gifts	150
arcade10	Arcade Starter	Play 10 Arcade+ games	150
arcade50	Arcade Ace	Play 50 Arcade+ games	500
maxpay1000	Four-Figure Payout	Receive a 1,000+ payout	300
wager10000	Volume Player	Wager 10,000 lifetime	300
cup10	Cup Legend	Win 10 Series Cups	600
trny5	Tournament Royalty	Win 5 tournaments	750
collector25	Wardrobe Vault	Own 25 paid cosmetics	750
level30	Master Level	Reach level 30	1000
level40	Mythic Level	Reach level 40	1500
level50	Level Summit	Reach level 50	2500
transfer10	Community Banker	Complete 10 player transfers	300
streak15	Untouchable	Reach a 15-win streak	1000
catalog50	Catalog Veteran	Play 50 catalog matches	500
clan5	Clan Loyalist	Play 5 clan games	300
allrounder	All-Rounder	Play Catalog, Series, Tournament, Friend and Arcade modes 500	

8.5 Levels 2–50

Level	Default	BONUS	Additional milestone
2	100	—	
3	150	—	
4	200	—	
5	250	—	
6	300	—	
7	350	—	
8	400	—	
9	450	—	
10	500	—	
11	550	—	
12	600	—	
13	650	—	
14	700	—	
15	750	—	
16	800	—	
17	850	—	
18	900	—	
19	950	—	
20	1000	—	
21	1050	—	
22	1100	—	
23	1150	—	
24	1200	—	
25	1250	—	
26	1300	—	
27	1350	—	
28	1400	—	
29	1450	—	
30	1500	Rare cosmetic	
31	1550	—	
32	1600	—	
33	1650	—	
34	1700	—	
35	1750	—	
36	1800	—	
37	1850	—	
38	1900	—	
39	1950	—	
40	2000	Epic cosmetic	
41	2050	—	
42	2100	—	
43	2150	—	
44	2200	—	
45	2250	—	
46	2300	—	
47	2350	—	
48	2400	—	
49	2450	—	
50	2500	Legendary cosmetic	

Level formula: floor(sqrt(XP/60)) + 1. All crossed levels are paid when one XP award skips levels. Prestige at level 10 resets level/XP, adds one rank, grants Rainbow frame and permanently adds +5% XP per rank.

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9. Shop inventory — 134 items

Category	ID	Name	Price	MAIN	Rarity	VIP restriction
skins	classic	Classic Gold	0		free	No
skins	platinum-vip	Platinum VIP	0		epic	VIP 4
skins	diamond-vip	Diamond VIP	0		legendary	VIP 5
skins	silver	Silver	25		common	No
skins	bronze	Bronze	25		common	No
skins	neon	Neon	50		uncommon	No
skins	arctic	Arctic	50		uncommon	No
skins	ember	Ember	75		uncommon	No
skins	ruby	Ruby	100		rare	No
skins	emerald	Emerald	100		rare	No
skins	sapphire	Sapphire	100		rare	No
skins	bitcoin	Bitcoin	150		rare	No
skins	koi	Koi Fish	150		rare	No
skins	galaxy	Galaxy	300		epic	No
skins	rainbow	Rainbow	500		legendary	No
skins	blackhole	Black Hole	600		legendary	No
skins	obsidian	Obsidian	250		rare	No
skins	solar	Solar Flare	450		epic	No
skins	frostfire	Frostfire	750		legendary	No
skins	quantum	Quantum Shift	1200		mythic	No
skins	jade	Jade Fortune	350		epic	No
skins	lunar	Lunar Halo	650		legendary	No
skins	legend	Legend Prism	0		mythic	VIP 8
skins	eternal	Eternal Flame	0		mythic	VIP 8
flags	in	India	25		common	No
flags	br	Brazil	25		common	No
flags	gb	UK	25		common	No
flags	us	USA	50		uncommon	No
flags	jp	Japan	50		uncommon	No
flags	de	Germany	50		uncommon	No
flags	ca	Canada	100		rare	No

flags	ae	UAE	100	rare	No
flags	au	Australia	100	rare	No
flags	it	Italy	100	rare	No
flags	kr	South Korea	200	epic	No
flags	earth	Earth	500	legendary	No
flags	sg	Singapore	100	rare	No
flags	za	South Africa	100	rare	No
flags	nz	New Zealand	150	rare	No
flags	pt	Portugal	150	rare	No
avatars	hero	Hero	25	common	No
avatars	ninja	Ninja	25	common	No
avatars	cop	Officer	50	uncommon	No
avatars	wizard	Wizard	50	uncommon	No
avatars	astro	Astronaut	100	rare	No
avatars	robot	Robot	100	rare	No
avatars	mask	Mask	100	rare	No
avatars	ghost	Ghost	200	epic	No
avatars	royal	Royal	500	legendary	No
avatars	agent	Agent	400	legendary	No
avatars	fairy	Fairy	1000	mythic	No
avatars	dragon	Dragon	800	legendary	No
avatars	alien	Alien	300	epic	No
avatars	fox	Fox	200	rare	No
avatars	panda	Panda	250	rare	No
avatars	unicorn	Unicorn	700	legendary	No
frames	none	None	0	free	No
frames	thin	Thin Ring	25	common	No
frames	double	Double Ring	50	uncommon	No
frames	gold	Gold Ring	100	rare	No
frames	fire	Fire Ring	150	rare	No
frames	ice	Ice Ring	200	epic	No
frames	neon	Neon Pulse	250	epic	No
frames	blackdiamond	Black Diamond Animated	0	legendary	VIP 6
frames	rainbow	Rainbow Spin	600	legendary	No
frames	emerald-ring	Emerald Orbit	200	rare	No
frames	cosmic-ring	Cosmic Halo	400	epic	No
frames	crown-ring	Crown Aura	700	legendary	No
colours	default	Default	0	free	No
colours	blue	Blue	50	uncommon	No
colours	green	Green	50	uncommon	No
colours	orange	Orange	100	rare	No
colours	purple	Purple	100	rare	No
colours	gold	Gold	200	epic	No
colours	pink	Pink	100	rare	No
colours	cyan	Cyan	100	rare	No
colours	red	Crimson	150	rare	No
colours	white	Pearl	250	epic	No
colours	rainbow	Rainbow	500	legendary	No
colours	glitch	Glitch	1000	mythic	No
fx	confetti	Standard Confetti	0	free	No
fx	coins	Gold Coins Fall	50	common	No
fx	fireworks	Fireworks	100	uncommon	No
fx	star	Star Burst	150	rare	No
fx	laser	Laser Show	300	epic	No
fx	money	Money Rain	300	epic	No
fx	phoenix	Phoenix Rise	600	legendary	No
fx	cherry	Cherry Blossom	250	epic	No
fx	thunder	Thunder Storm	450	epic	No
fx	aurora-burst	Aurora Burst	550	legendary	No
fx	trophy	Trophy Shower	700	legendary	No
fx	dragon	Dragon Roar	1200	mythic	No
themes	midnight	Midnight	0	free	No
themes	ocean	Ocean	50	common	No
themes	forest	Forest	100	uncommon	No
themes	sunset	Sunset	150	rare	No
themes	space	Space	300	epic	No
themes	casino	Casino Gold	300	epic	No
themes	aurora	Aurora	600	legendary	No
themes	desert	Desert Gold	200	rare	No
themes	icepalace	Ice Palace	350	epic	No
themes	royalhall	Royal Hall	700	legendary	No
themes	lava	Lava Chamber	850	legendary	No
themes	cyber	Neon Cyberpunk	1000	mythic	No
sounds	standard	Standard	0	free	No
sounds	8bit	8-Bit Retro	50	common	No
sounds	orch	Orchestral	100	uncommon	No
sounds	dj	DJ Mix	200	rare	No
sounds	trailer	Movie Trailer	300	epic	No
sounds	synth	Synthwave	250	epic	No
sounds	crystal	Crystal Chime	350	epic	No
sounds	bass	Deep Bass	450	legendary	No
sounds	choir	Epic Choir	500	legendary	No
emojis	fire	Fire	25	common	No
emojis	party	Party	25	common	No
emojis	down	Thumbs down	25	common	No
emojis	drool	Drooling	50	uncommon	No
emojis	poop	Poop	50	uncommon	No
emojis	angry	Angry	50	uncommon	No
emojis	think	Thinking	100	rare	No
emojis	explode	Exploding head	100	rare	No
emojis	nerd	Nerdy	100	rare	No
emojis	money	Money face	200	epic	No
emojis	crown	Crown	200	epic	No
emojis	ctrl	Controller	200	epic	No
emojis	gem	Diamond	400	legendary	No

emojis	slot	Slot	400	legendary	No
emojis	hundo	100	500	legendary	No
emojis	rocket	Rocket	150	rare	No
emojis	trophy	Trophy	250	epic	No
emojis	robot	Robot	150	rare	No
emojis	alien	Alien	250	epic	No
emojis	fox	Fox	100	rare	No
emojis	panda	Panda	100	rare	No

Purchases use MAIN, apply current VIP shop discount where eligible, update house shop/sink accounting and persist ownership. Bots may purchase affordable non-VIP inventory. E1 crates and E6 crafting also grant items through the same ownership model.

10. Economy+ and coin utility

Code Feature		Exact current configuration and accounting boundary
E1	Mystery Crates	Common 75 MAIN, Rare 200, Epic 500, Legendary 1,000. Common/Rare grant 1–3 items; Epic/Legendary grant 2–3; selected tier is the rarity floor. Duplicate results remain possible.
E2	Trading Post	Rotating bot listings and player listings. Player listing minimum is 25. Buyer pays the listing price, seller receives 90%, and house records a rounded 10% fee.
E3	Staking Vault	Stake/unstake minimum input 100. Completed weeks pay 1% of staked balance; each claim is capped at 500 and recorded as house-funded reward. Demo can advance seven days.
E4	Subscriptions	30-day Plus 300 MAIN, Pro 700, Elite 1,500. Plus: badge + monthly cosmetic. Pro: exclusive-table entitlement + cosmetic + 2× quest rewards. Elite: Pro perks + Cyber theme + one-hour XP-booster bundle. Repurchase extends from the later of now/current expiry.
E5	Coin Boosters	2× XP for one hour costs 200 MAIN; +5 percentage points rakeback for 24 hours costs 300 MAIN, capped at 25% effective. Repurchase extends active duration.
E6	Cosmetic Crafting	Uncommon-floor recipe 150 MAIN; Rare-floor 400; Epic-floor 800. One random item at/above the floor is granted; duplicates are possible; last 50 crafts retained.
E7	Event Ticket Packs	1 ticket = 100 MAIN; 3 = 250; 7 = 500. Tickets persist as non-payout demo utility; last 50 purchases retained.
E8	Clan Treasury	Minimum contribution 50 MAIN; control defaults to 100. Funds are non-withdrawable, grant no wagering edge and count as sink/social resource—not house revenue. Utility level = min(10, 1 + floor(treasury/1,000)).
E9	Private Room Upgrades	Basic 0, Neon 200 MAIN, Royal 500, Cosmic 1,000. Persistent visual tier is applied to stored Private Rooms; it changes no odds, payout, queue or matching behavior; last 20 purchases retained.

Purchases use MAIN and retain Economy history. Ordinary shop, crates, subscriptions, boosters, crafting, tickets and room upgrades use shop/sink accounting. E8 is deliberately excluded from house revenue. E7 makes no payout promise. All values are play-coin demo values and are Admin/runtime configuration candidates for a server implementation.

11. Bots and automatic activity

Area	Behavior
Roster	99 initial bots; automatic creation default 1 every 15 seconds up to 250; Admin bounds 99–1,000, interval 5–3,600 seconds, batch 1–10
Games	Eight Coin Toss and eight Catalog runs per normal cycle; Series, tournaments and all 20 Arcade+ modes
Social	Friend requests, bot-to-bot friendships, lobby chat, room/clan/gift activity logs
Economy	Shop purchases, Trading Post, transfers, event participation and varied top-ups
Top-up	Triggers below Admin-configured default 500; base 400–1,600; first eligible top-up +50% if promotion active; not house revenue
Player count	Home, separate Player Directory, separate Leaderboard and Admin use persisted roster length; production must separate bots from humans
Turbo	Player header provides 1×/20×/100×/500×/1,000× stress speeds; normal saved default 1×; Turbo changes simulation volume only.

12. Player screens and navigation

Tab	Purpose
home	Home dashboard and customization
lobby	Live bets and activity
play	Coin Toss
series	Player-vs-bot and unattended bot-vs-bot Series Cups; bot-filled tournaments; double-or-nothing rooms
games	33-game P2P Catalog
newgames	20-game Arcade+
leaderboard	Ranked list
players	Separate Player Directory
community	Social Hub
progressionplus	Progress+
economyplus	Economy+ and Coin Utility
shop	134-item shop
season	VIP, quests, achievements
history	Nine history categories
updates	Release notes
wallet	Wallet and quick RG
stats	Advanced statistics
verify	Fairness verifier
services	UX, Platform, API, Security, RG, Statements, Offers

Navigation includes desktop sidebar search, grouped quick jump, Ctrl/Command+K and / command palette, Escape close, mobile dock, exact-game command entries, Catalog/Arcade search/category/Favorites/jump, and query-parameter deep links.

13. Histories and statistics

History category	Included records
All Games	Coin Toss, Catalog, Arcade and feature games
P2P Catalog	Per-game picks, result, fee, proof and carry
Friend Challenges	Opponent/result/payout
Private Rooms	Room lifecycle and room games
Clan Games	Matches and clan tournaments

Arcade+	G1–G20
Progress+	Claims, Prestige and pass activity
Economy+	Crates, trade, staking, subscriptions, boosters, E6–E9
Social	Friends, gifting and communication actions

Statistics include games/wins/losses/draws/carries, streaks, net, biggest stake, lifetime wagered, total/max payout, fees, average stake/payout, return on wagered, game-family participation, Cups/tournaments, friend/room/clan and Arcade counts, plus recent payout trend.

14. Admin screens and controls

Admin destination Controls	
Overview	KPIs, alerts, quick actions, revenue/jackpot/RNG
Live Operations	Maintenance, bots/growth, player, queue, transfers, top-ups
Feature Hub	Social/Arcade/Progress/Economy and coin-utility telemetry, including 33 Catalog/20 Arcade destinations and Catalog earnings paths
Feature Directory	101 records with search/category/status and quick links
Rates & Jackpot	Fees, jackpot and universal rules
Economy	P&L, taps/sinks, simulator and compensation
Promotions	First-top-up toggle, scheduled campaigns, broadcast and claims
VIP & Levels	Eight tiers and 49 paginated level rewards
Tournaments	Create/filter/sort brackets
Audit & Data	Audit, proofs, histories and exports
Trust Center	Analytics, anti-cheat, RG, PWA/API/notification/2FA/statement posture

Admin data over 20 rows uses filter/sort/pagination. Admin cannot end durable Player self-exclusion. Exact Player game links are available from Feature Directory and command palette.

15. Platform, security, RG and UX

Code	Feature	Status	Current detail
RG-D	Demo RG Controls	Implemented	Persistent deposit/session limits, cool-off, durable self-exclusion, loss limit, reality graph, bank and Auto Bet stop.
T1	Installable PWA	Implemented	Manifest, 192/512 icons, install readiness and offline service-worker cache.
T2	Multi-language	Implemented	Persistent English, Hindi, Bengali, Tamil and Telugu primary navigation/Home translations.
T3	Public API	Partial	Working in-browser API explorer, key rotation, OpenAPI spec and request log; server API still required.
T4	Real-time Analytics	Implemented	Five-second local player/game/revenue/queue/social/Arcade samples, KPIs, chart and export.
T5	Push Notifications	Partial	Browser permission, preferences, local notifications and history; remote push backend still required.
SEC1	Two-Factor Authentication	Partial	Web Crypto six-digit TOTP enrollment, verification and verified disable demo; account backend required.
SEC2	Anti-Cheat Detection	Implemented	Rules scan for impossible balances, extreme win rate, duplicate proof, malformed payout and Turbo risk.
SEC3	Public Fairness Page	Implemented	Standalone proof verification without account access.
SEC4	Monthly Activity Statement	Implemented	Monthly preview with wager, payout, fee, P/L and top-up totals plus JSON/CSV downloads and email simulation.
RG1	Deposit Limits	Implemented	Persistent daily/weekly/monthly limits, immediate decreases, delayed increases and enforced top-up checks.
RG2	Session Time Limits	Implemented	Persistent maximum duration with enforced cool-off before new games.
RG3	Durable Self-Exclusion	Implemented	Timed or permanent shared-state exclusion with no early Player/Admin undo.
RG4	Reality Check Graph	Implemented	Configurable reminders, retained session P/L points, graph and explicit up/down callout.
UX1	Accessibility Center	Implemented	Persistent high contrast, reduced motion, text scale, colour-vision presets and screen-reader hints.
UX2	Customizable Dashboard	Implemented	Player-selected and reordered Home KPI cards and sections with reset and persistence.
UX3	Saved Bet & Game Presets	Implemented	Up to 20 named Coin/Catalog/Arcade presets that fill controls without auto-wagering and retain RG checks.
UX4	Smart Game Discovery	Implemented	Explainable favorite, variety, current-game, daily and collection recommendations with exact links.

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16. Complete Feature Directory — 101 records

Category	Code	Feature	Status	Description	Destination
Core Games	CORE-1	Coin Toss	Implemented	Escrowed HEADS/TAILS P2P match with fee, jackpot, proof, manual take and auto-match.	Player play
Core Games	CORE-2	P2P Catalog	Implemented	Thirty-three proof-driven games with per-game queues, manual take, carries and history.	Player games
Core Games	CORE-3	Series Cups	Implemented	Bo3, Bo5, Bo7 and Advantage Cups for player/bot or bot-only play.	Player series
Core Games	CORE-4	Tournaments	Implemented	Single-flip and Bo3 brackets with bot fill, VIP discount and 75/25 prizes.	Player series · Admin trny
Core Games	CORE-5	Jackpot	Implemented	Fee-funded, threshold-armed byte-00 jackpot with rollover.	Player play · Admin rates
Core Games	CORE-6	Fairness Verifier	Implemented	Local SHA-256 recomputation for retained Coin and Catalog proof inputs.	Player verify
Social & Community	S1	Friends & Buddies	Implemented	Friend list, bot requests and connections, online status, direct challenge and friend-first bets.	Player community/friends
Social & Community	S2	Private Rooms	Implemented	Invite/friend access, custom game and stake rules, details and history.	Player community/rooms
Social & Community	S3	Player Profiles	Implemented	Stats, VIP, skill, cosmetics, achievements, form graph and recent games.	Player community/profile
Social & Community	S4	Spectator Mode	Implemented	Animated viewer with participants, picks, pot, fee, result and proof.	Player community/spectate
Social & Community	S5	Gifting	Implemented	MAIN coin and paid cosmetic gifts with fee, minimum, cap and history. Text, owned emojis, autonomous bot	Player community/gifts

Social & Community	S6	Lobby Chat	Implemented	conversation, replies, mute/block and sent history.	Player community/chat
Social & Community	S7	Clans & Teams	Implemented	Clan identity, roster, team matches, tournaments, leaderboard and history.	Player community/clans
Arcade+ Games	G1	Lucky Wheel	Implemented	Daily free and paid spins with coin and cosmetic prizes.	Player newgames/wheel
Arcade+ Games	G2	Scratch Cards	Implemented	Three card tiers, nine reveals and match-based payouts.	Player newgames/scratch
Arcade+ Games	G3	Dice Roll	Implemented	Exact double 30× or low/high range 1.8×.	Player newgames/dice
Arcade+ Games	G4	Weekly Raffle	Implemented	Player/bot tickets with 80% winner payout and 20% house share.	Player newgames/raffle
Arcade+ Games	G5	Multiplier Ladder P2P	Implemented	Matched rung-or-bust P2P duel with split handling.	Player newgames/ladder
Arcade+ Games	G6	War Card Game	Implemented	2--Ace cards with automatic War rounds and P2P settlement.	Player newgames/war
Core Games	CAT18	Rock Paper Scissors Duel	Implemented	Two hidden simultaneous picks; standard RPS winner takes the post-fee P2P pot.	Player games/rps
Core Games	CAT19	Closest to 21	Implemented	Each player receives proof-derived cards; closest total to 21 without exceeding it wins.	Player games/closest21
Core Games	CAT20	Triple Coin Majority	Implemented	Players select HEADS or TAILS and three proof flips decide the majority side.	Player games/triplecoin
Core Games	CAT21	Sequence Builder	Implemented	Players choose different two-symbol starters and extend a proof sequence toward a target pattern.	Player games/sequencebuilder
Core Games	CAT22	Dice Sum Duel	Implemented	Each participant receives two proof-derived dice; the higher sum wins and equal sums split.	Player games/dicesumduel
Core Games	CAT23	Colour Spectrum Duel	Implemented	Players claim different colour bands mapped across byte 0-255; result inside a claimed band wins.	Player games/colourspectrum
Core Games	CAT24	Prime vs Composite	Implemented	Opposite picks resolve from a number between 2 and 251; primality determines the winner.	Player games/primecomposite
Core Games	CAT25	Median Number Battle	Implemented	Two player picks and one proof number form three values; the player closest to the median wins.	Player games/medianbattle
Core Games	CAT26	Streak Survivor	Implemented	Opposite sides race through proof flips; the first side to achieve a four-result streak wins.	Player games/streaksurvivor
Core Games	CAT27	Territory Capture	Implemented	Players claim non-overlapping map sectors and successive proof bytes capture territory until majority.	Player games/territory
Core Games	CAT28	Modulo Four Duel	Implemented	Two players claim different remainders from 0-3; a proof byte modulo four decides, while unclaimed results carry.	Player games/modulo4
Core Games	CAT29	Poker High Duel	Implemented	Five proof-derived cards per entrant; standard hand rank determines the P2P winner.	Player games/pokerhigh
Core Games	CAT30	Three Dice Poker	Implemented	Three proof dice each; triple beats pair, then total/high die, with exact ties split.	Player games/threedicepoker
Core Games	CAT31	Last Digit Duel	Implemented	Different 0-9 picks compete against a proof number last digit; unclaimed digits carry.	Player games/lastdigit
Core Games	CAT32	Binary Code Duel	Implemented	Different three-bit codes compete by Hamming distance to the proof code; equal distance splits.	Player games/binaryduel
Core Games	CAT33	Coin Balance Battle	Implemented	Distinct predictions compete on the number of HEADS in ten proof flips; equal distance splits.	Player games/coinbalance
Arcade+ Games	G7	Plinko Drop	Implemented	Drop a chip through a proof-derived peg path into prize multiplier slots.	Player newgames/plinko
Arcade+ Games	G8	Mini Slots	Implemented	Three proof-derived reels with transparent symbol odds, paylines and capped prizes.	Player newgames/slots
Arcade+ Games	G9	Quick Keno	Implemented	Pick numbers from a compact board and compare them with proof-derived draws.	Player newgames/keno
Arcade+ Games	G10	Bingo Rush	Implemented	Fast 3×3 card with proof draws; complete a line before the draw limit.	Player newgames/bingo
Arcade+ Games	G11	Treasure Hunt	Implemented	Choose tiles on a hidden map containing coins, multipliers, keys and traps.	Player newgames/treasure
Arcade+ Games	G12	Memory Match	Implemented	Resolve eight proof-driven memory moves and earn published pair-count prizes.	Player newgames/memory
Arcade+ Games	G13	Drop Ball	Implemented	Release a ball into columns with proof-derived bounce direction and visible multiplier pockets.	Player newgames/dropball
Arcade+ Games	G14	Daily Trivia	Implemented	One proof-retained daily question attempt; a correct answer pays the published 3× reward.	Player newgames/trivia
Arcade+ Games	G15	Fishing Reel	Implemented	Cast, wait and reel proof-derived fish rarities for collection and coin rewards.	Player newgames/fishing
Arcade+ Games	G16	Penalty Shootout	Implemented	Pick shot direction against a proof-derived goalkeeper for a five-kick score challenge.	Player newgames/penalty
Arcade+ Games	G17	Coin Pusher	Implemented	Five proof-derived drops push stacks toward published 0×-6× reward trays.	Player newgames/coinpusher
Arcade+ Games	G18	Tower Builder	Implemented	Choose floor 3, 5 or 7 and survive every proof-derived risk check for the published multiplier.	Player newgames/tower
Arcade+ Games	G19	Match-3 Rush	Implemented	A proof-derived 5×5 gem board pays by horizontal and vertical three-symbol matches.	Player newgames/match3
Arcade+ Games	G20	Mystery Vault	Implemented	Choose one of five keys; the proof-selected winning key pays 4.5×.	Player newgames/vault
Progression	P1	Battle Pass	Implemented	Monthly free/premium XP milestones and reward claims.	Player progressionplus/pass
Progression	P2	7-Day Login Calendar	Implemented	Escalating automatic rewards with day-7 cosmetic and BONUS.	Player progressionplus/calendar
Progression	P3	Weekly Challenges	Implemented	Wins, game variety and streak goals with larger rewards.	Player progressionplus/weekly
Progression	P4	Prestige	Implemented	Level reset for permanent XP boost, badge and Rainbow frame.	Player progressionplus/prestige
Progression	P5	Skill Matchmaking	Implemented	Five skill tiers and preferred peer-bot matching with fallback.	Player progressionplus/skill
Economy+	E1	Mystery Crates	Implemented	Four tiers with 1-3 cosmetics and a guaranteed rarity floor.	Player economyplus/crates
Economy+	E2	Trading Post	Implemented	Bot listings, player sales and a 10% house	Player economyplus/trade

				transaction fee.	
Economy+	E3	Staking Vault	Implemented	Flexible staking with completed-week 1% interest and claim cap.	Player economyplus/staking
Economy+	E4	Subscriptions	Implemented	Plus, Pro and Elite 30-day tiers with engagement perks.	Player economyplus/subscription
Economy+	E5	Coin Boosters	Implemented	Time-limited 2× XP or +5% rakeback consumables.	Player economyplus/boosters
Economy+	E6	Cosmetic Crafting	Implemented	Spend MAIN on uncommon, rare or epic rarity-floor cosmetic recipes with retained history.	Player economyplus/utility
Economy+	E7	Event Ticket Packs	Implemented	Buy persistent non-payout demo utility tickets in 1, 3 or 7-ticket packs.	Player economyplus/utility
Economy+	E8	Clan Treasury	Implemented	Non-withdrawable MAIN contributions create clan utility resources and levels without wagering advantage.	Player economyplus/utility
Economy+	E9	Private Room Upgrades	Implemented	Unlock persistent Basic, Neon, Royal or Cosmic visual room tiers with no gameplay advantage.	Player economyplus/utility
Player & Retention	RET-1	8-Tier VIP	Implemented	Monthly wager tiers, rakeback, permanent rewards, discounts and priority.	Player season · Admin vip
Player & Retention	RET-2	Achievements & Levels	Implemented	Forty-nine achievements, configurable rewards through level 50 and milestone cosmetics.	Player season · Admin vip
Player & Retention	RET-3	Player History	Implemented	Nine categories, search, sorting, pagination, details and export.	Player history
Player & Retention	RET-4	Advanced Statistics	Implemented	Lifetime wager, total/max payout, fees, averages, ROI and payout trend.	Player stats
Wallet & Commerce	WAL-1	Segmented Wallet	Implemented	MAIN, BONUS, REFERRAL, RAKEBACK and BANK with stake caps.	Player wallet
Wallet & Commerce	WAL-2	Top-up Promotion	Implemented	Admin-controlled first top-up bonus for player and first bot top-ups.	Player wallet · Admin promo
Wallet & Commerce	WAL-3	Shop & Cosmetics	Implemented	Nine expanded categories, 38 new items, VIP rewards, discounts and equip controls.	Player shop
Wallet & Commerce	WAL-4	Transfers & Bot Economy	Implemented	Active bot games/social/Arcade/shop/transfers, configurable 500-balance top-ups and automatic roster growth.	Player wallet · Admin ops
Operations	OPS-1	Command Center	Implemented	KPIs, alerts, quick actions, revenue, jackpot and RNG monitoring.	· Admin dash
Operations	OPS-2	Feature Hub	Implemented	B1–B4 telemetry and feature administration controls.	· Admin features
Operations	OPS-3	Audit & Exports	Implemented	Audit trail, review flags, filtering, pagination and data exports.	· Admin audit
Operations	OPS-4	Promotions Manager	Implemented	Scheduled campaigns now support Player credit/cash-drop claims, next-top-up activation, one-claim tracking and Admin counts.	Player services/offers · Admin promo
Responsible Gaming	RG-D	Demo RG Controls	Implemented	Persistent deposit/session limits, cool-off, durable self-exclusion, loss limit, reality graph, bank and Auto Bet stop.	Player services/rg · Admin trust
Platform	T1	Installable PWA	Implemented	Manifest, 192/512 icons, install readiness and offline service-worker cache.	Player services/platform · Admin trust
Platform	T2	Multi-language	Implemented	Persistent English, Hindi, Bengali, Tamil and Telugu primary navigation/Home translations.	Player services/platform · Admin trust
Platform	T3	Public API	Partial	Working in-browser API explorer, key rotation, OpenAPI spec and request log; server API still required.	Player services/api · Admin trust
Platform	T4	Real-time Analytics	Implemented	Five-second local player/game/revenue/queue/social/Arcade samples, KPIs, chart and export.	· Admin trust
Platform	T5	Push Notifications	Partial	Browser permission, preferences, local notifications and history; remote push backend still required.	Player services/platform · Admin trust
Security	SEC1	Two-Factor Authentication	Partial	Web Crypto six-digit TOTP enrollment, verification and verified disable demo; account backend required.	Player services/security · Admin trust
Security	SEC2	Anti-Cheat Detection	Implemented	Rules scan for impossible balances, extreme win rate, duplicate proof, malformed payout and Turbo risk.	Player services/security · Admin trust
Security	SEC3	Public Fairness Page	Implemented	Standalone proof verification without account access.	Player verify
Security	SEC4	Monthly Activity Statement	Implemented	Monthly preview with wager, payout, fee, P/L and top-up totals plus JSON/CSV downloads and email simulation.	Player services/statements · Admin trust
Responsible Gaming	RG1	Deposit Limits	Implemented	Persistent daily/weekly/monthly limits, immediate decreases, delayed increases and enforced top-up checks.	Player services/rg · Admin trust
Responsible Gaming	RG2	Session Time Limits	Implemented	Persistent maximum duration with enforced cool-off before new games.	Player services/rg · Admin trust
Responsible Gaming	RG3	Durable Self-Exclusion	Implemented	Timed or permanent shared-state exclusion with no early Player/Admin undo.	Player services/rg · Admin trust
Responsible Gaming	RG4	Reality Check Graph	Implemented	Configurable reminders, retained session P/L points, graph and explicit up/down callout.	Player services/rg · Admin trust
UX & Accessibility	UX1	Accessibility Center	Implemented	Persistent high contrast, reduced motion, text scale, colour-vision presets and screen-reader hints.	Player services/ux
UX & Accessibility	UX2	Customizable Dashboard	Implemented	Player-selected and reordered Home KPI cards and sections with reset and persistence.	Player services/ux
UX & Accessibility	UX3	Saved Bet & Game Presets	Implemented	Up to 20 named Coin/Catalog/Arcade presets that fill controls without auto-wagering and retain RG checks.	Player services/ux
UX & Accessibility	UX4	Smart Game Discovery	Implemented	Explainable favorite, variety, current-game, daily and collection recommendations with exact links.	Player services/ux
Suggested — LiveOps & Social	LIVE1	Event Calendar & Scheduled Play	Suggested	Upcoming tournaments, Cups, promotions and reminders in one calendar with timezone support.	Roadmap item
Suggested — LiveOps & Social	LIVE2	Match Replay & Shareable Proof	Suggested	Step-by-step replay, share link and redacted proof package for games and tournaments.	Roadmap item
Suggested — LiveOps & Social	LIVE3	Clan Seasons & Cooperative Quests	Suggested	Season divisions, shared clan goals, contribution ledger, rewards and archived standings.	Roadmap item

Suggested — Trust & Compliance	TRUST1 Identity, Age & Jurisdiction Checks	Suggested	Age gate, KYC status, geofencing and jurisdiction-specific feature eligibility.	Roadmap item
Suggested — Trust & Compliance	TRUST2 Privacy & Data Rights Center	Suggested	Consent history, data export, correction and deletion requests with retention status.	Roadmap item
Suggested — Trust & Compliance	TRUST3 Dispute & Support Case Center	Suggested	Open cases from a game or transaction, attach proof, track SLA and record resolution.	Roadmap item
Suggested — Trust & Compliance	TRUST4 Device & Session Management	Suggested	View active devices, revoke sessions, detect unusual login locations and require step-up verification.	Roadmap item
Suggested — Operations	OP5 Status & Incident Center	Suggested	Service health, incident timeline, maintenance updates, postmortems and player-facing status notices.	Roadmap item

17. Accounting model

Flow	Accounting
Coin/Catalog	Fee removed from pot; jackpot contribution separated; house fee/net and sinks updated
Cup/Tournament	Configured rake recognized; prizes paid from entrant pool
Direct Arcade	MAIN spend enters shop/sink family; coin payout is tap/promotion cost
Raffle	Ticket value stays in pool; only 20% draw share is house revenue
Shop/Crafting/Tickets/Room upgrades	MAIN spend and shop/sink accounting; ownership/utility retained
Clan Treasury	MAIN removed to non-withdrawable social resource; sink only, no house revenue
Trading Post	Buyer pays; seller receives 90%; house 10%
Bot top-ups	Demo liquidity tap, varied values, promotion cost for first bonus; never deposit revenue
Rewards	BONUS/tap and promotion cost where house-funded
Admin comps	Cannot exceed available net revenue

18. Responsible Gaming

Control	Behavior
Loss limit	Blocks new wagering once session net reaches negative limit
Deposit limits	Daily/weekly/monthly; top-up enforcement; decreases immediate, increases/removal delayed in demo
Session limit	Maximum minutes triggers cool-off and blocks games
Self-exclusion	Timed/permanent persisted lock; no early Player/Admin undo
Reality check	Configurable reminders, session P/L points/graph and up/down callout
Bank	Park MAIN outside wagering balance
Auto Bet stop	Default –200, configurable –50 to –10,000

19. Production boundary and deferred roadmap

The demo is not approved for public networked, real-money or regulated use. Production requires server-authoritative ledger/RG/settlement, authenticated accounts/Admin RBAC/MFA, independent RNG commitment service, safe output rendering/CSP, durable database, certified mathematics, KYC/AML/geofencing, bot/human analytics separation, observability and independent audits.

Code	Suggested production feature
LIVE1	Event Calendar & Scheduled Play — Upcoming tournaments, Cups, promotions and reminders in one calendar with timezone support.
LIVE2	Match Replay & Shareable Proof — Step-by-step replay, share link and redacted proof package for games and tournaments.
LIVE3	Clan Seasons & Cooperative Quests — Season divisions, shared clan goals, contribution ledger, rewards and archived standings.
TRUST1	Identity, Age & Jurisdiction Checks — Age gate, KYC status, geofencing and jurisdiction-specific feature eligibility.
TRUST2	Privacy & Data Rights Center — Consent history, data export, correction and deletion requests with retention status.
TRUST3	Dispute & Support Case Center — Open cases from a game or transaction, attach proof, track SLA and record resolution.
TRUST4	Device & Session Management — View active devices, revoke sessions, detect unusual login locations and require step-up verification.
OP5	Status & Incident Center — Service health, incident timeline, maintenance updates, postmortems and player-facing status notices.

20. Validation baseline

Release validation must include Player/Admin syntax and boot, exact registry counts, deterministic game-rule checks, wallet/ledger conservation, bot balance invariants, Feature Directory total/unique/status checks, IDs, tab/panel mapping, CSS balance, DOCX ZIP/open, support-file parsing and live HTTP exact-content checks. The authoritative runtime sources are index.html and admin.html; this complete specification and Appendix V of the DOCX consolidate their current v10.4 rules. Earlier DOCX appendices remain historical release records and do not override Appendix V.

End of complete specification.

END OF APPENDIX V — TOSSMATCH v10.4 COMPLETE SPECIFICATION

TossMatch • Play-coins demonstration •